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(54) **INDUCTOR ON MICROELECTRONIC DIE**

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H01L 23/00 (2006.01)

(52) **U.S. Cl.**

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2924/19104 (2013.01)

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See application file for complete search history.

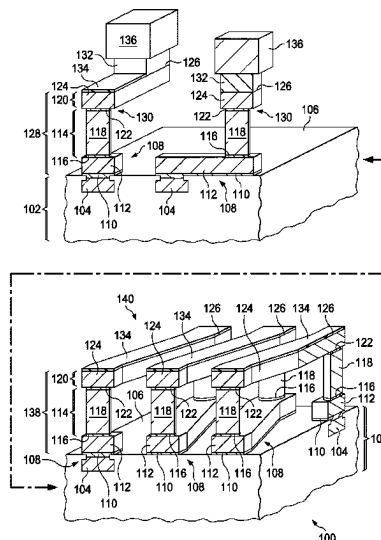
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20 Claims, 22 Drawing Sheets



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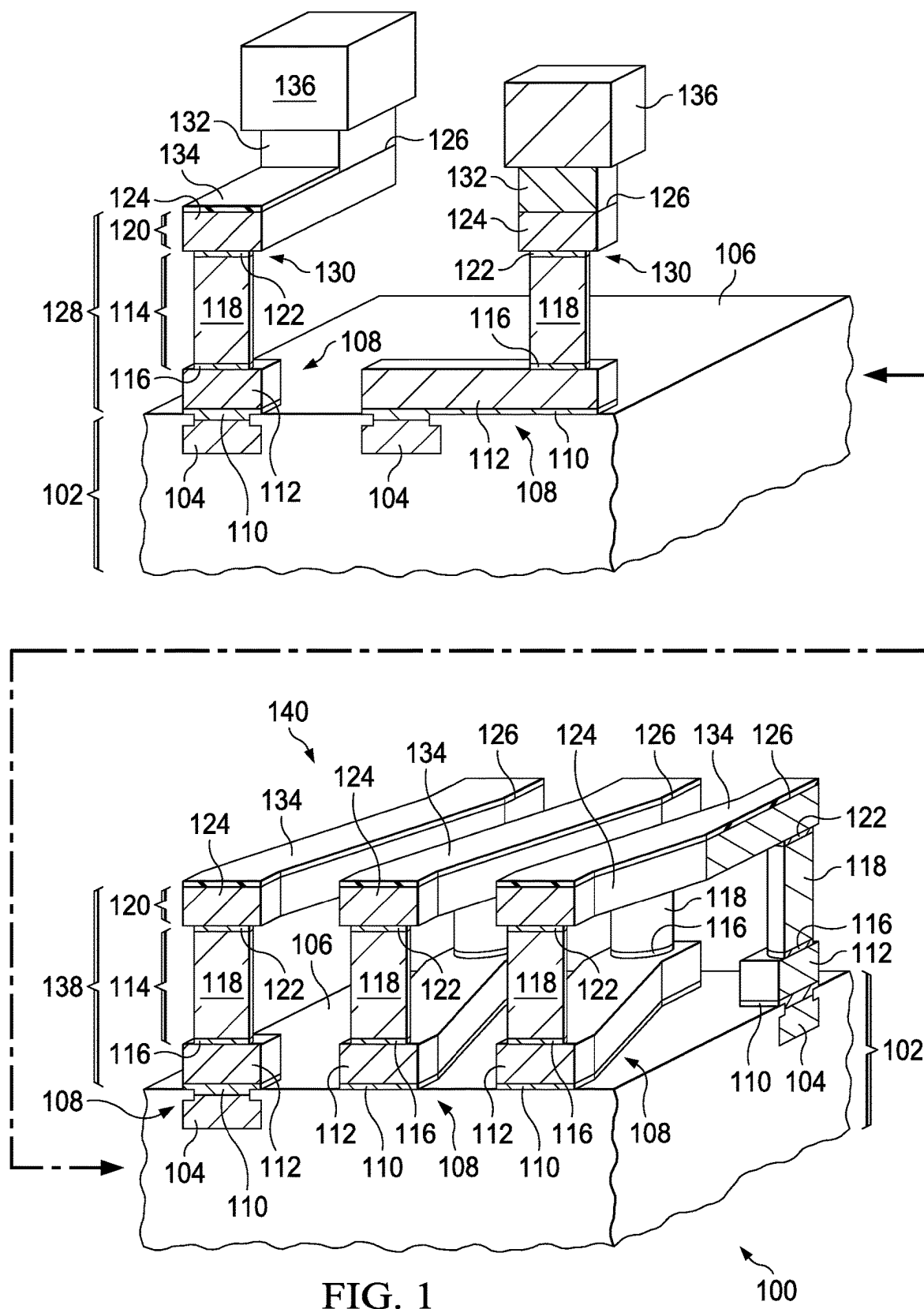
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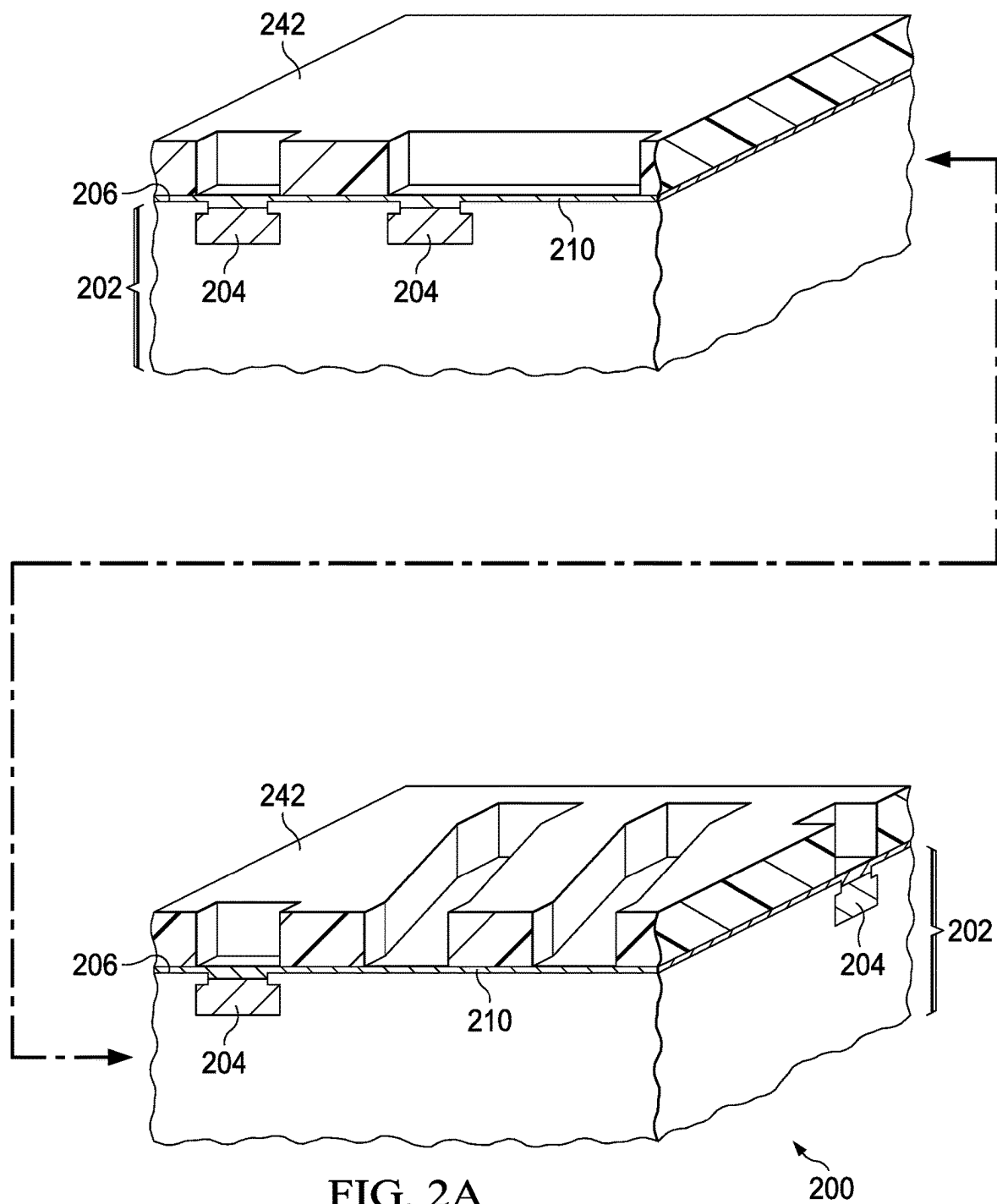
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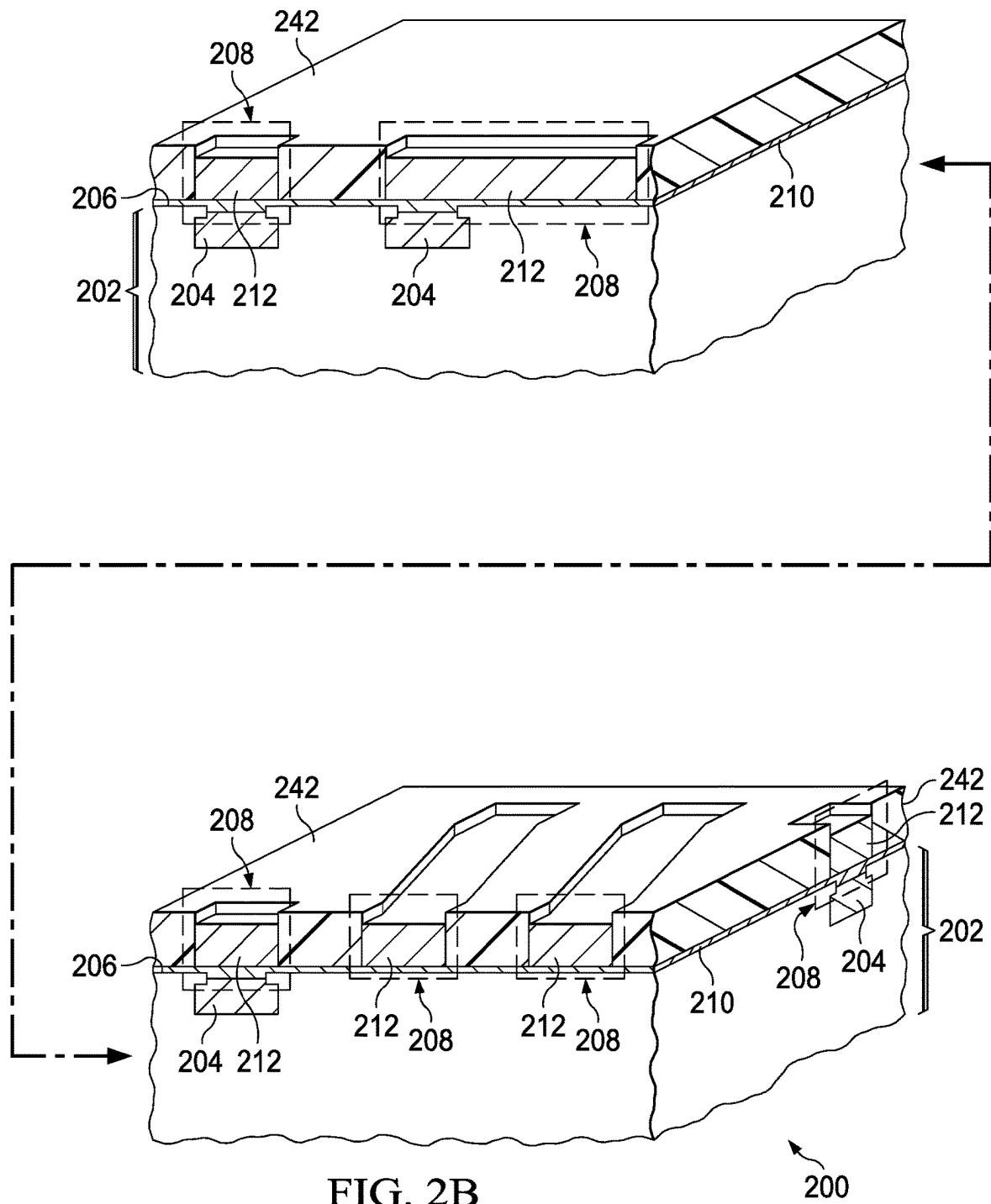
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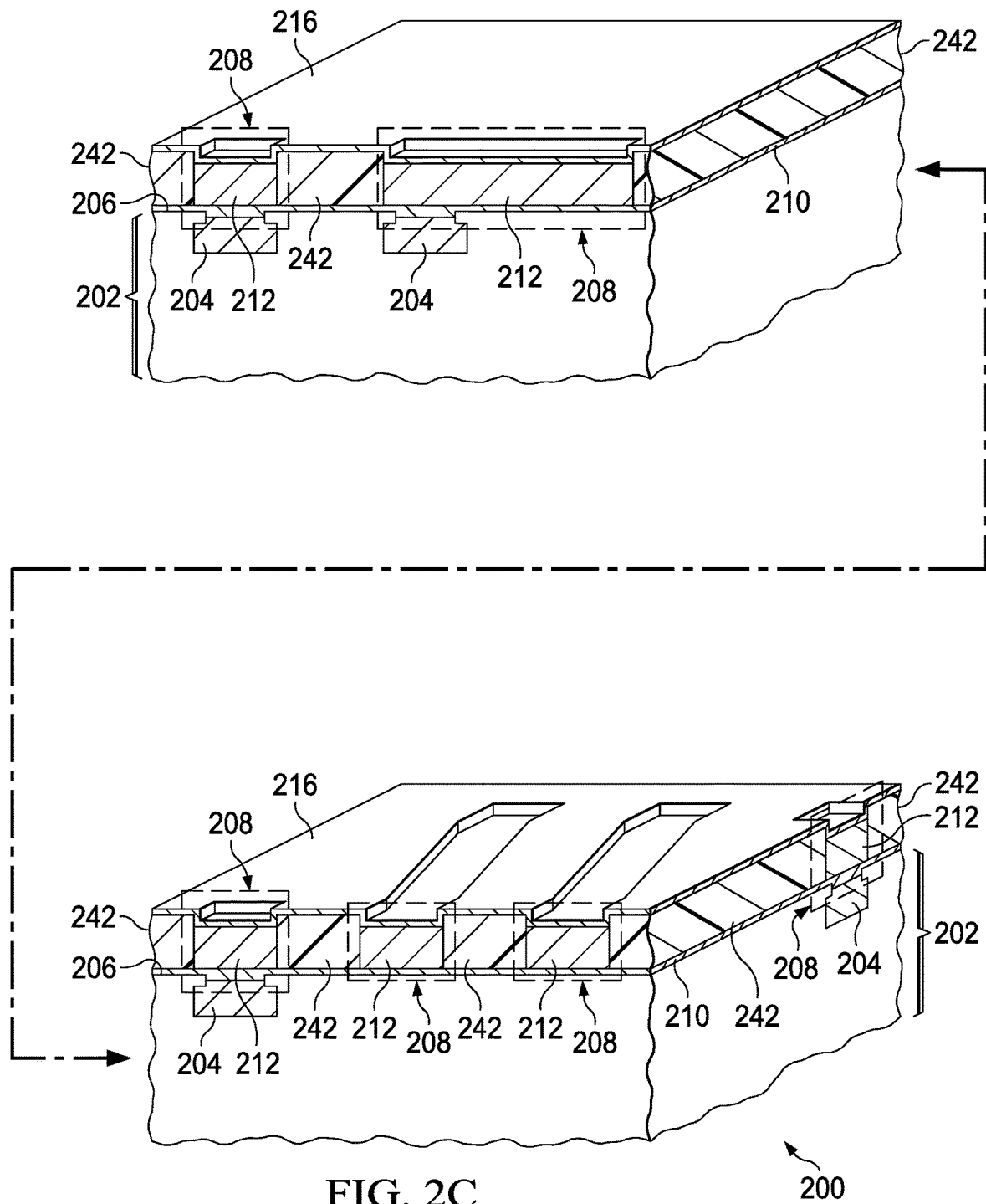
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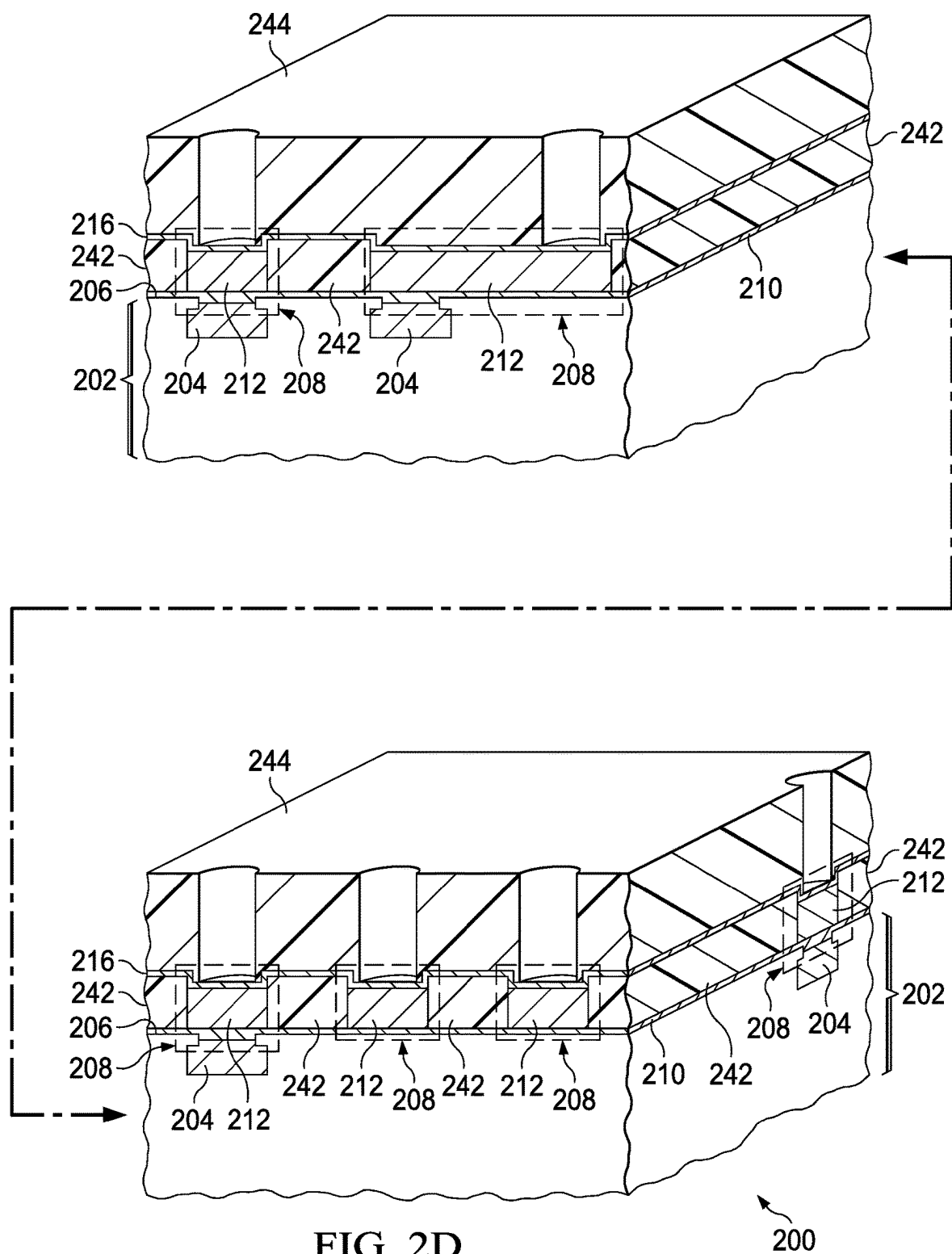
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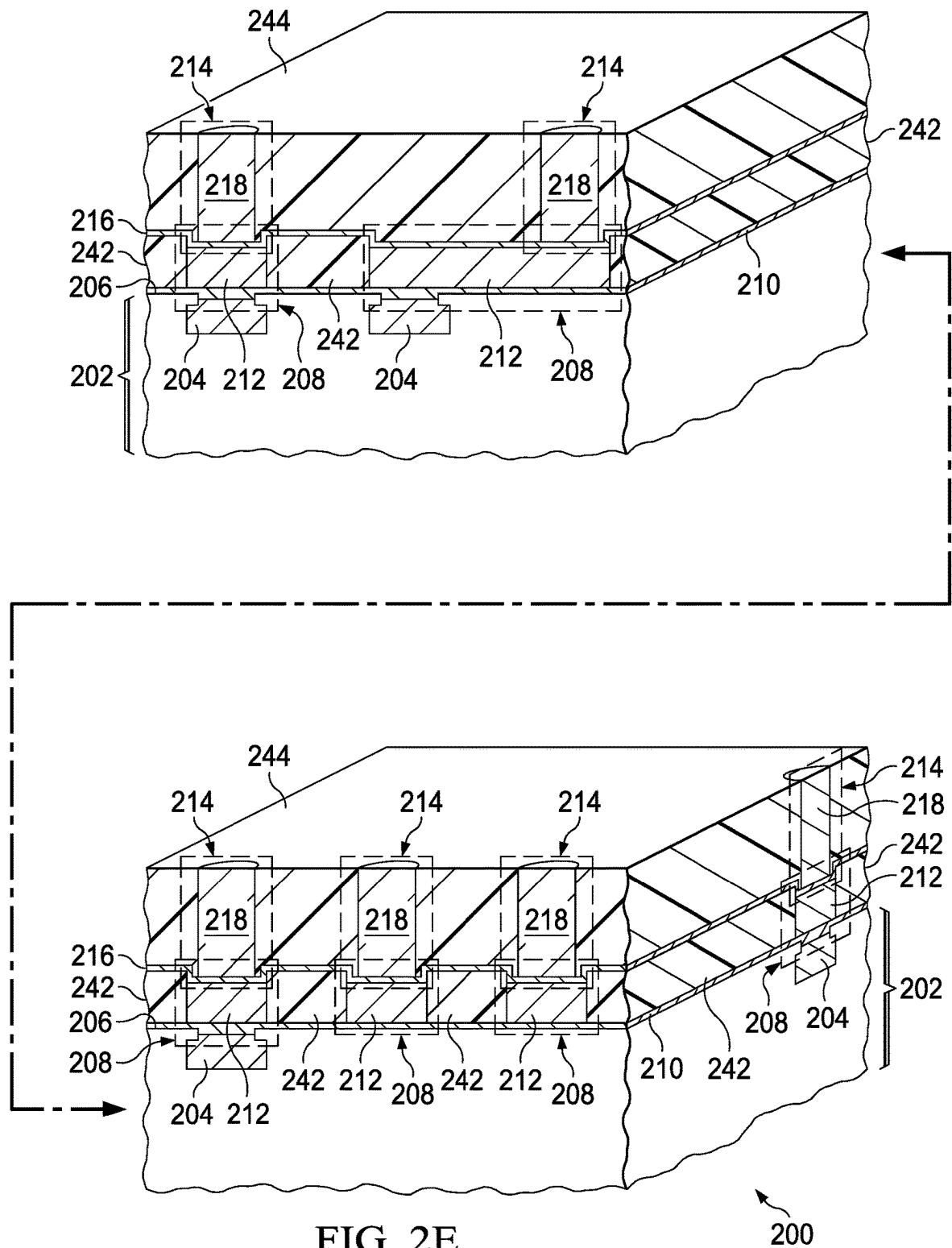


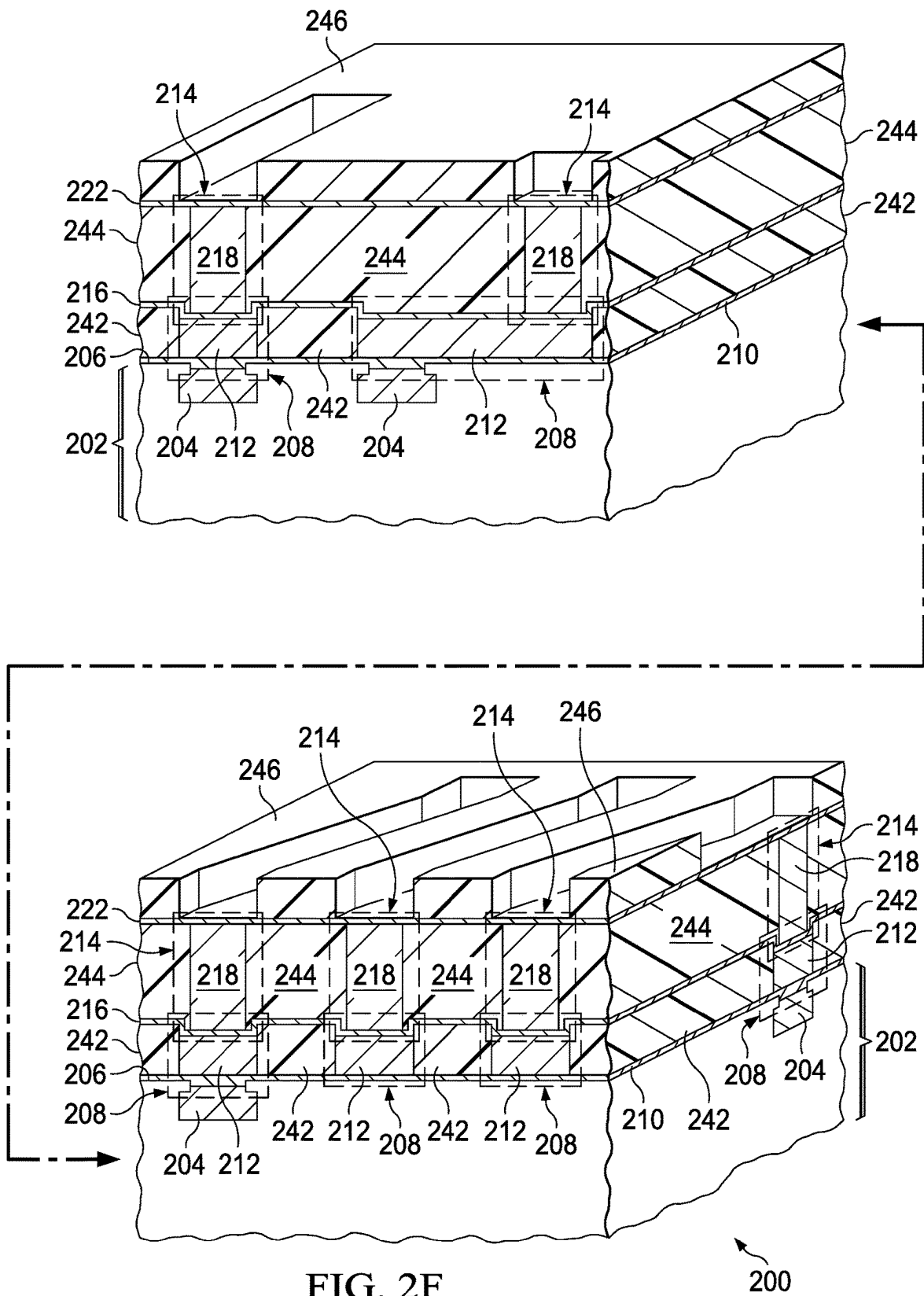


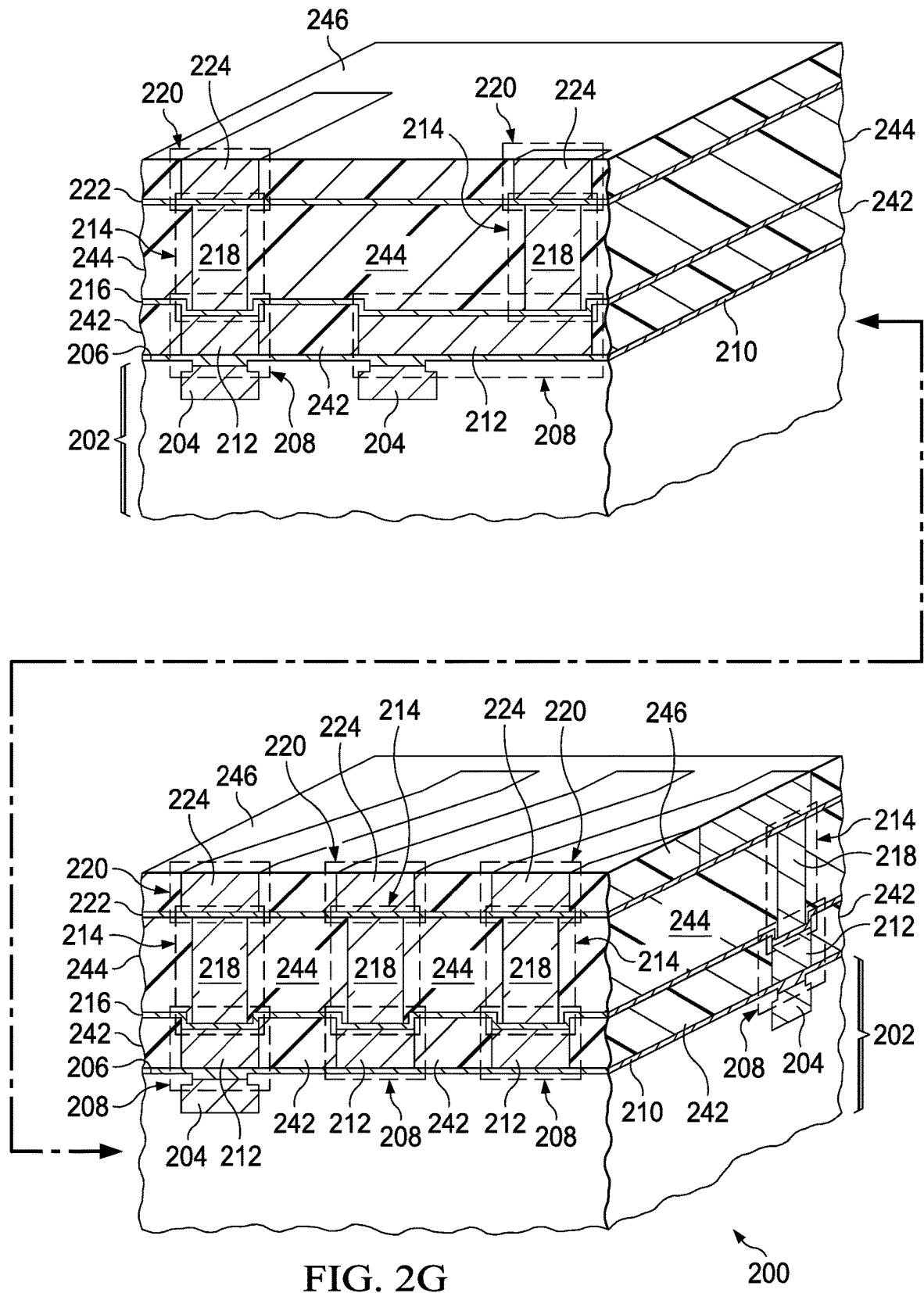


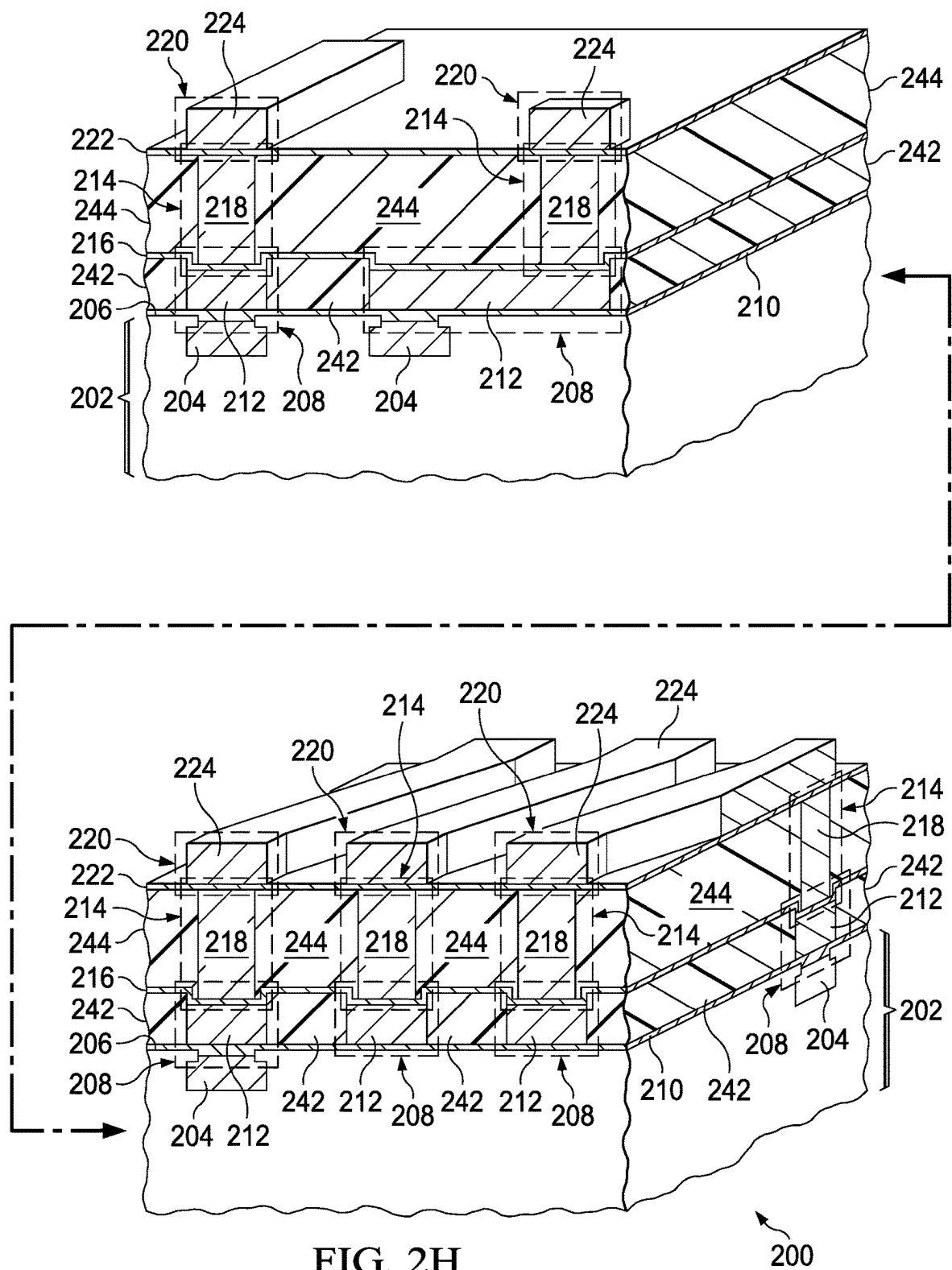


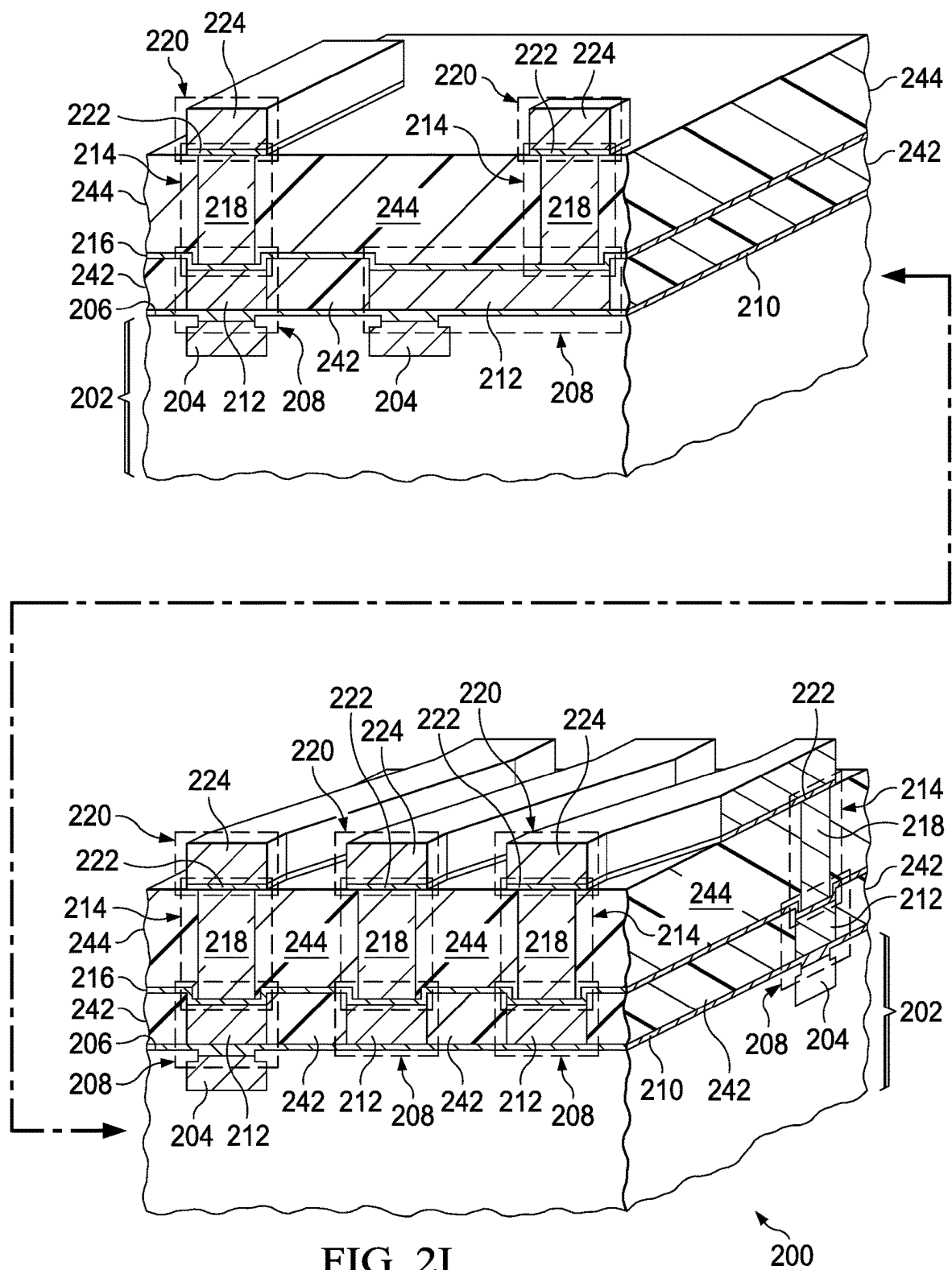


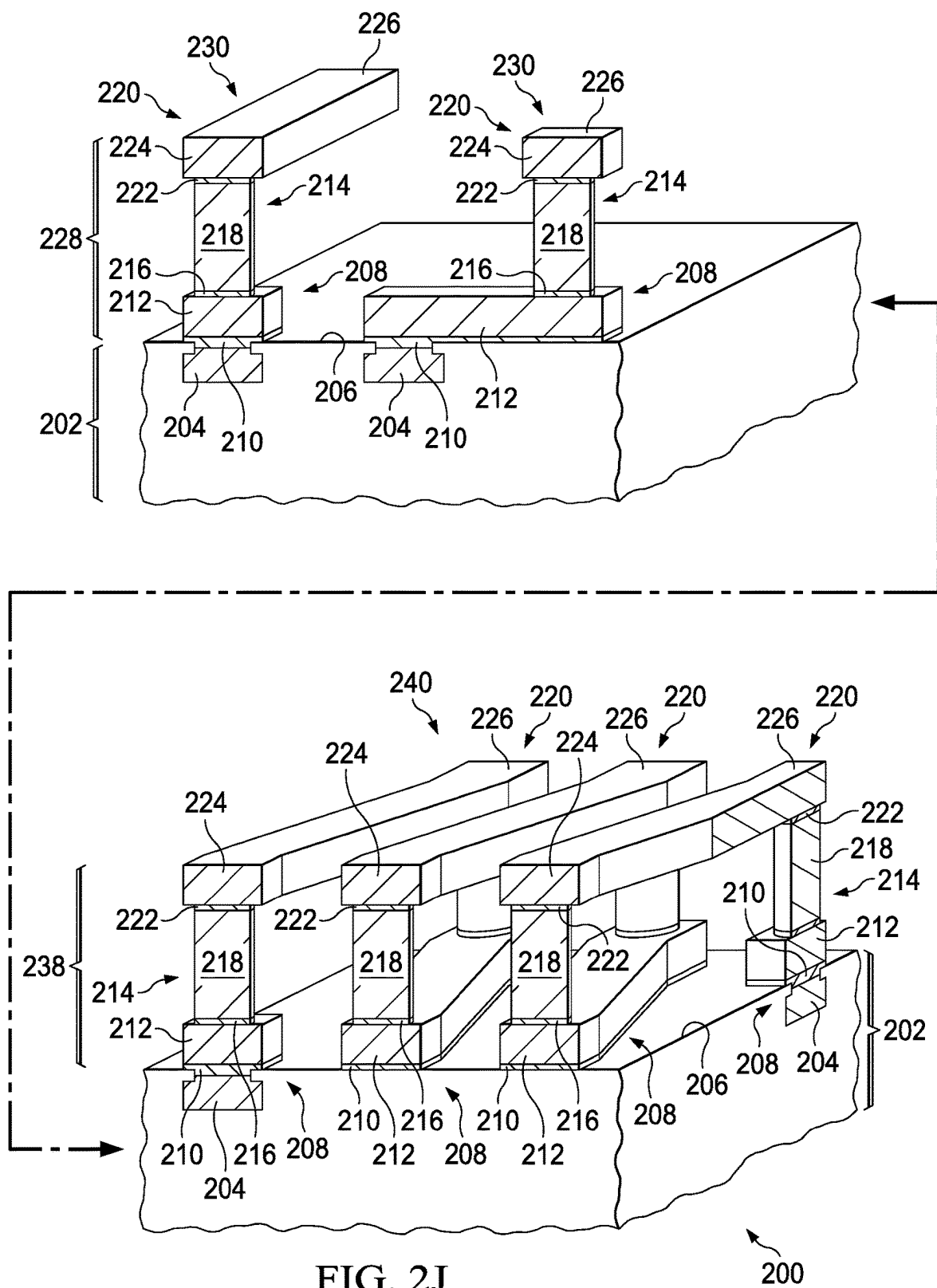


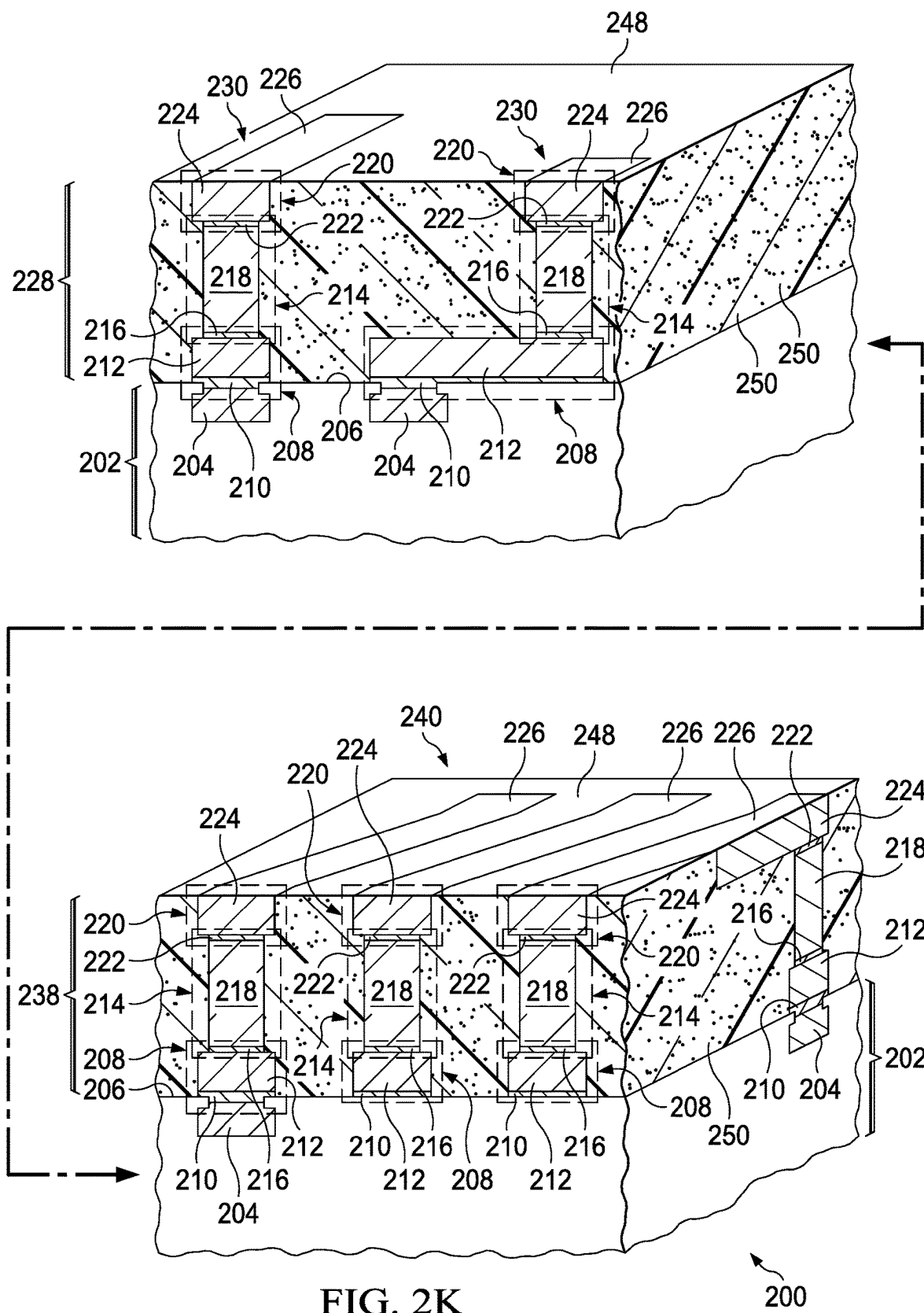


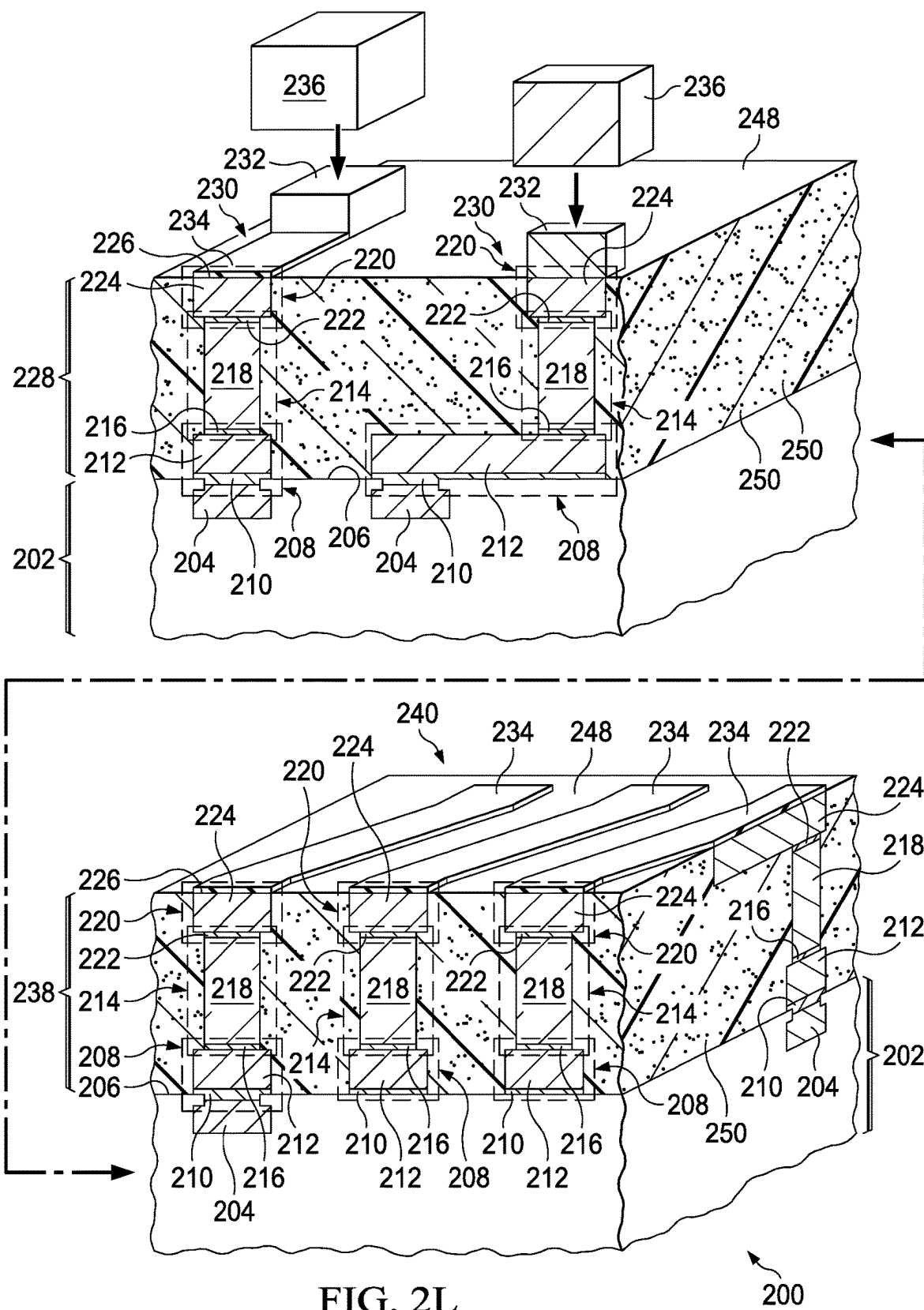


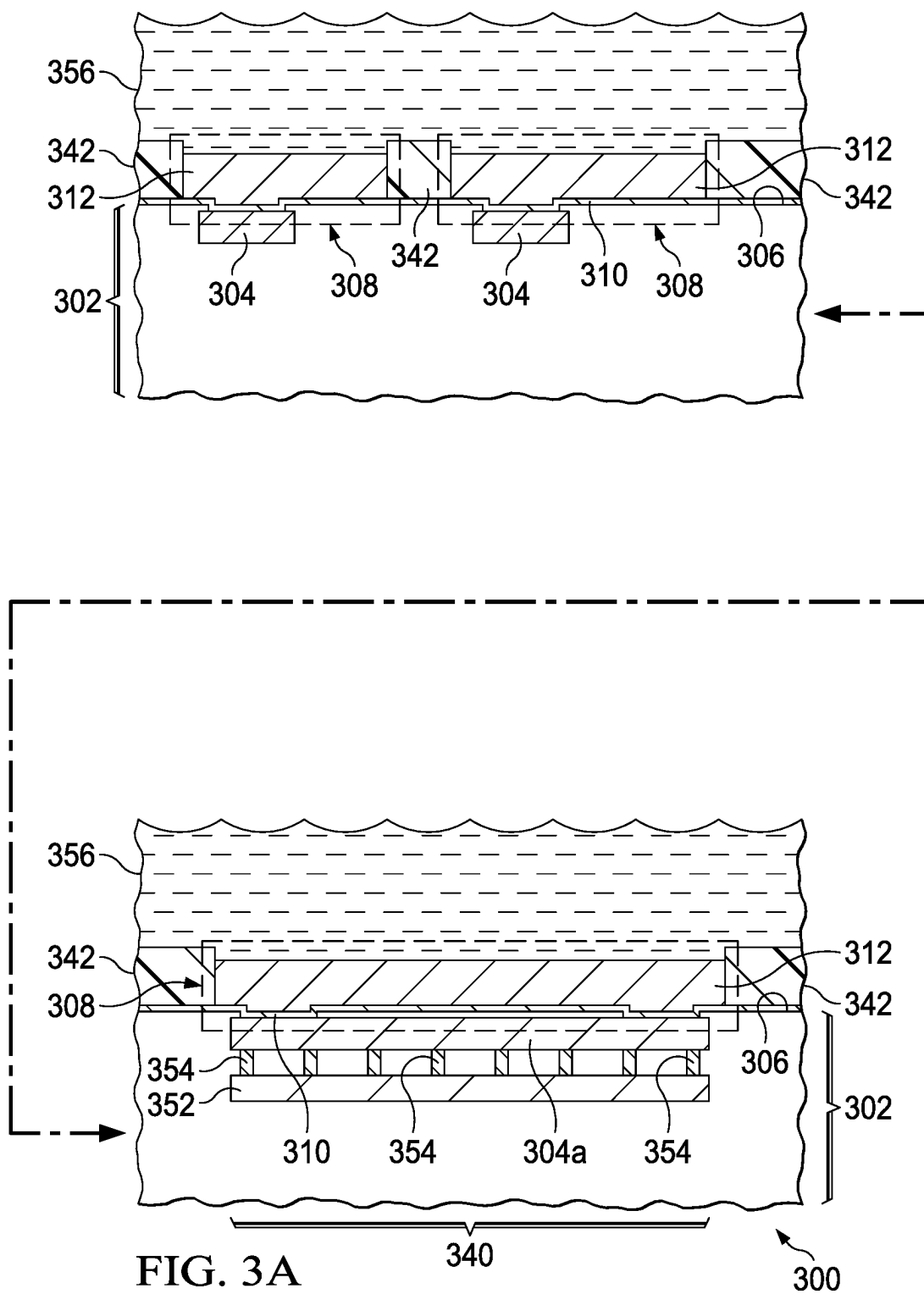


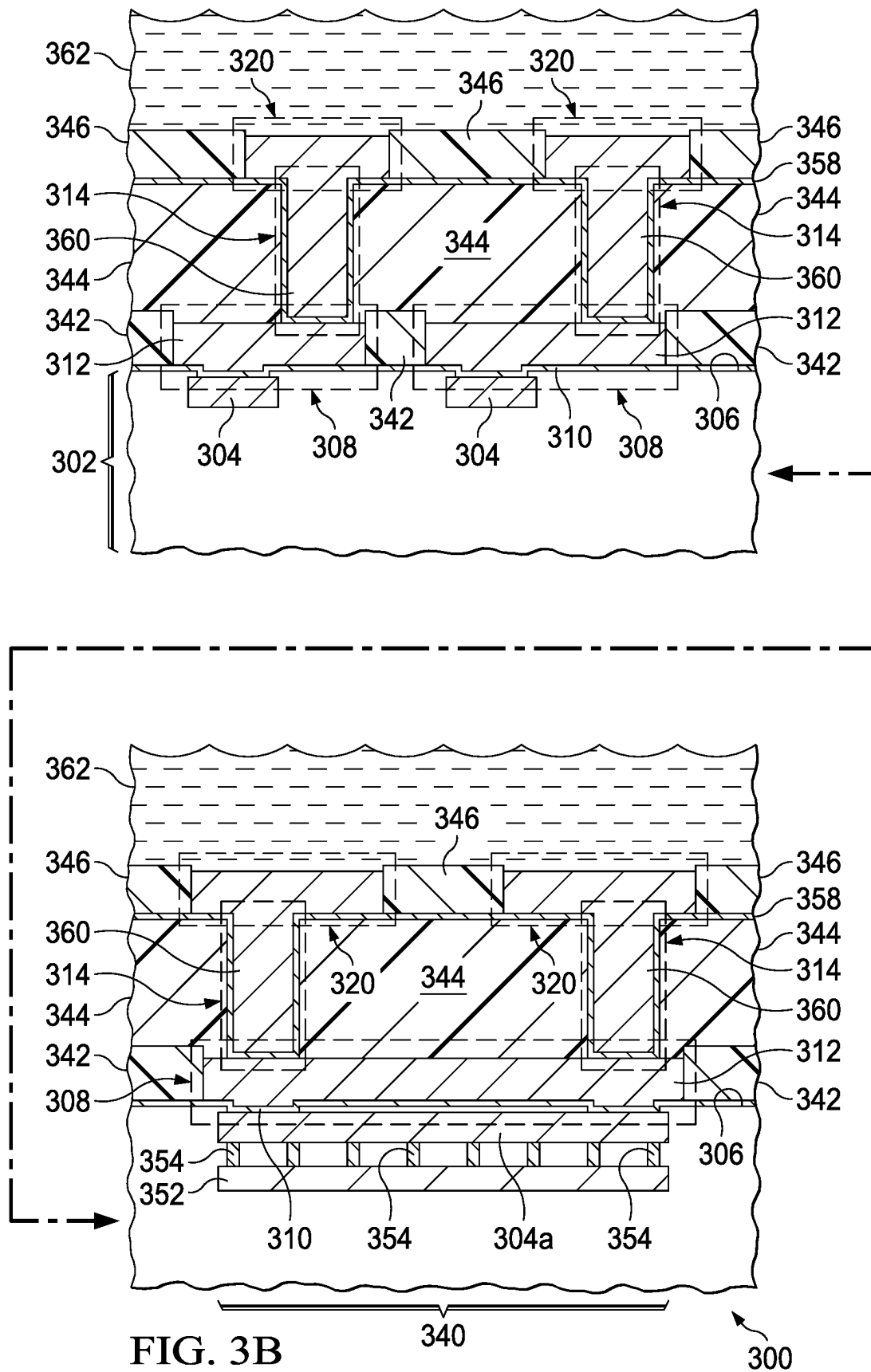


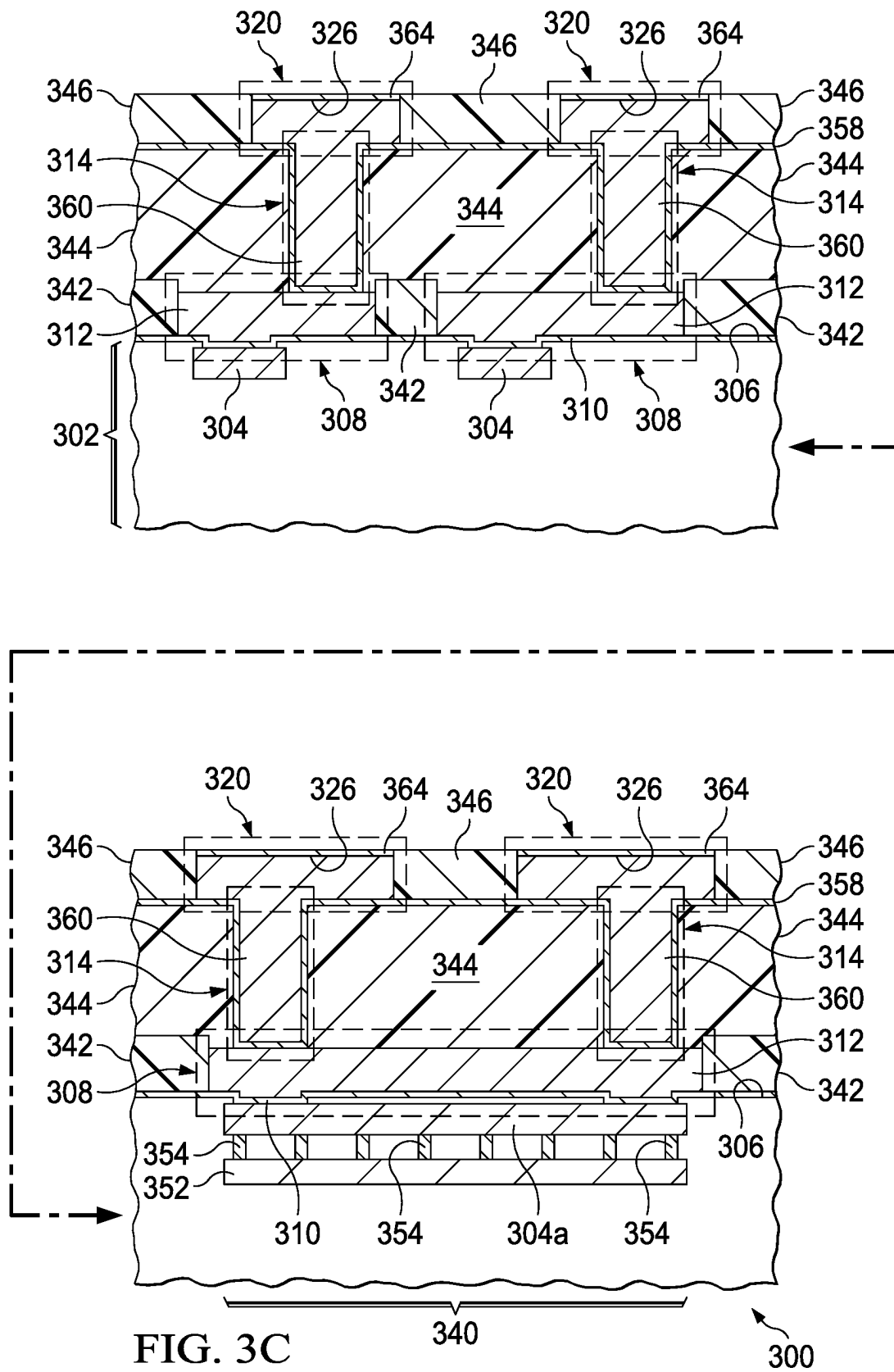


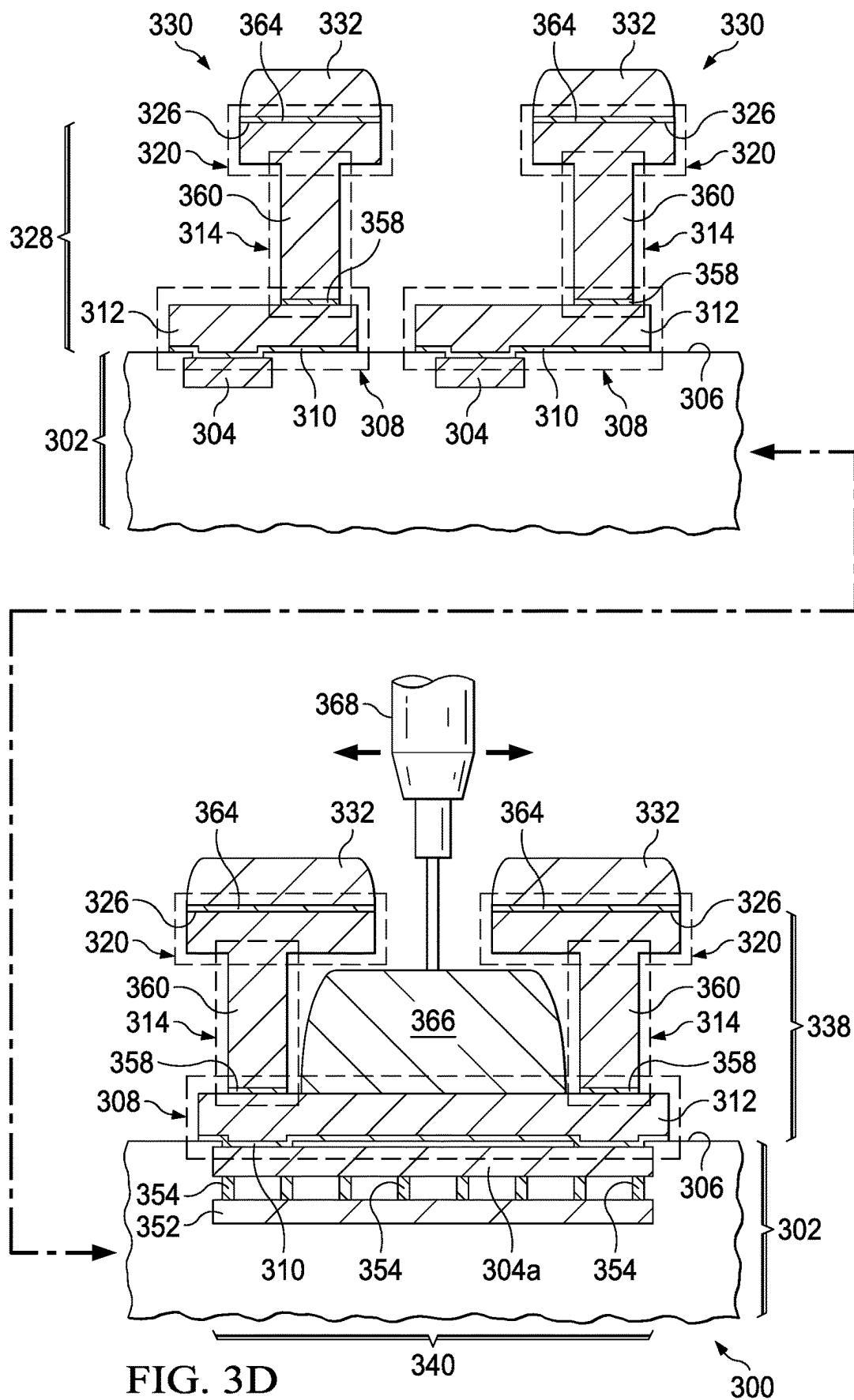


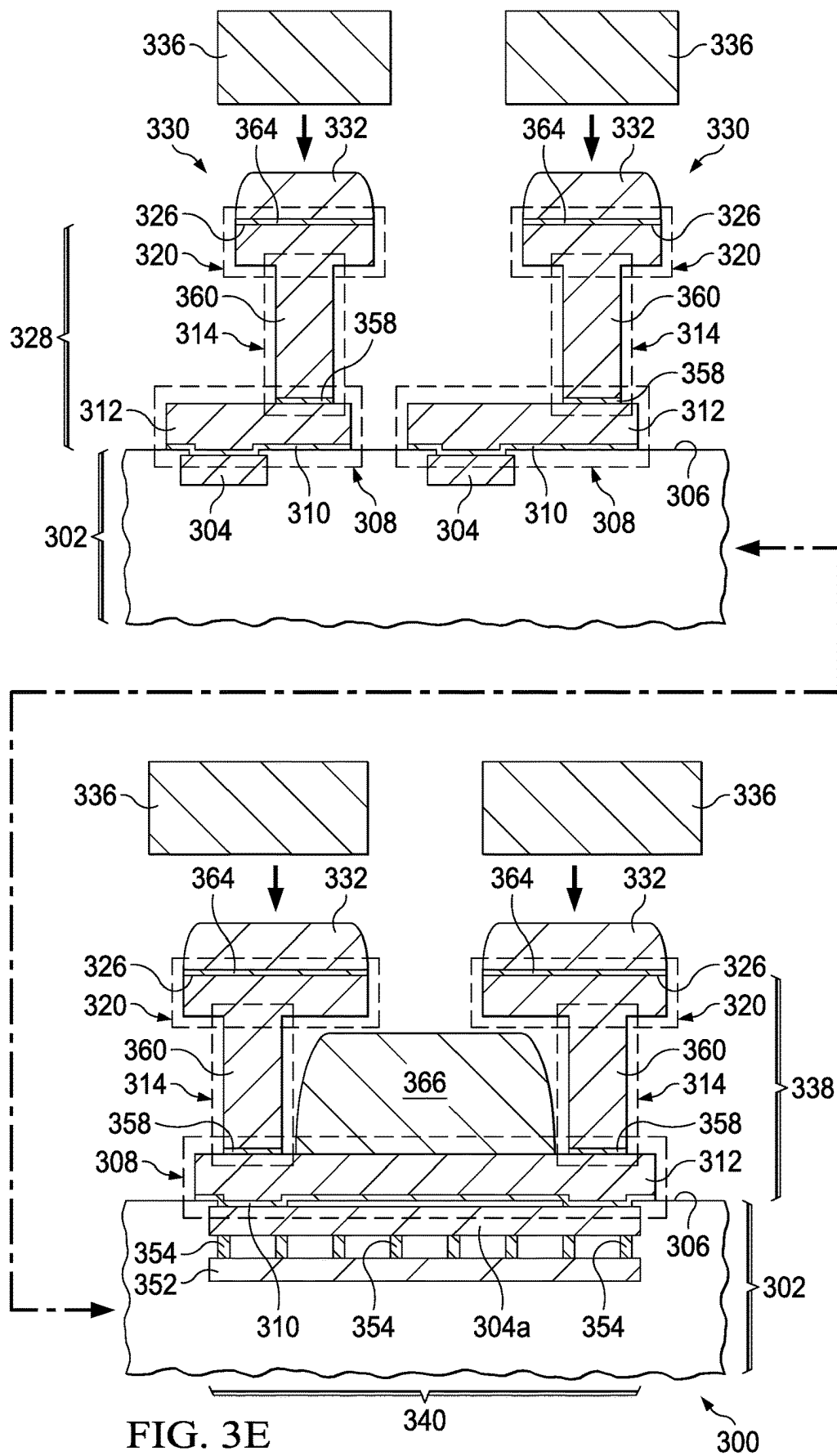


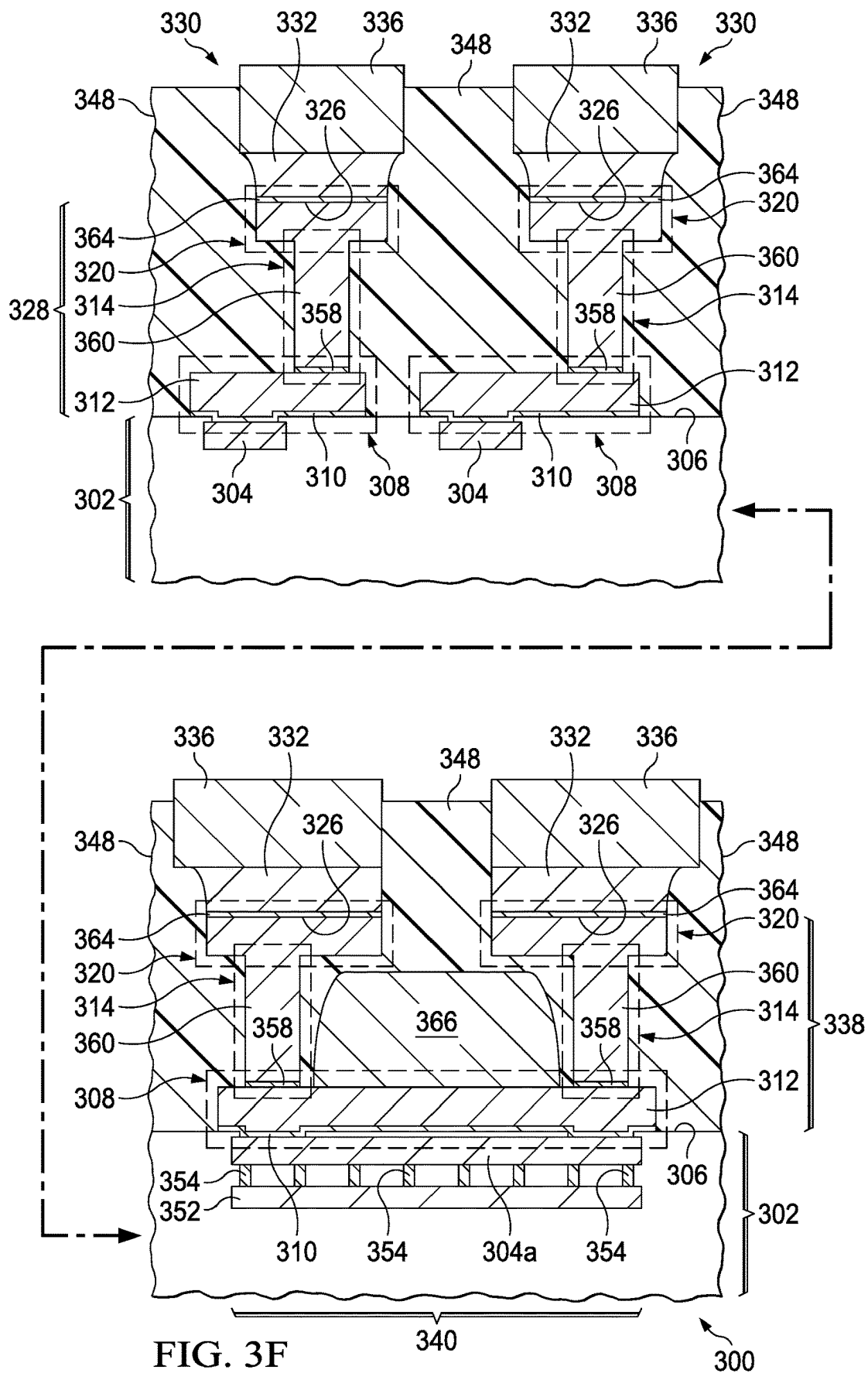


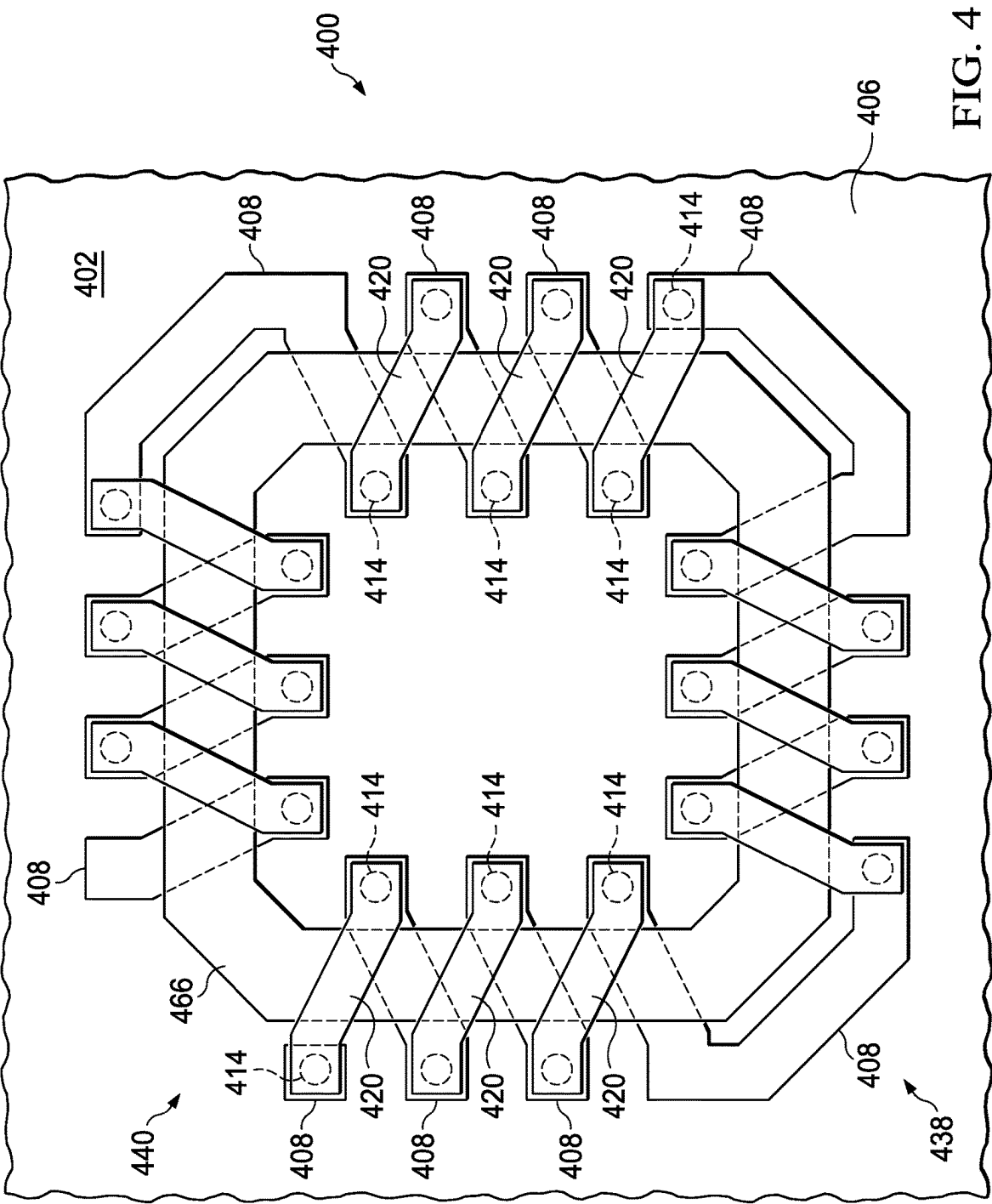












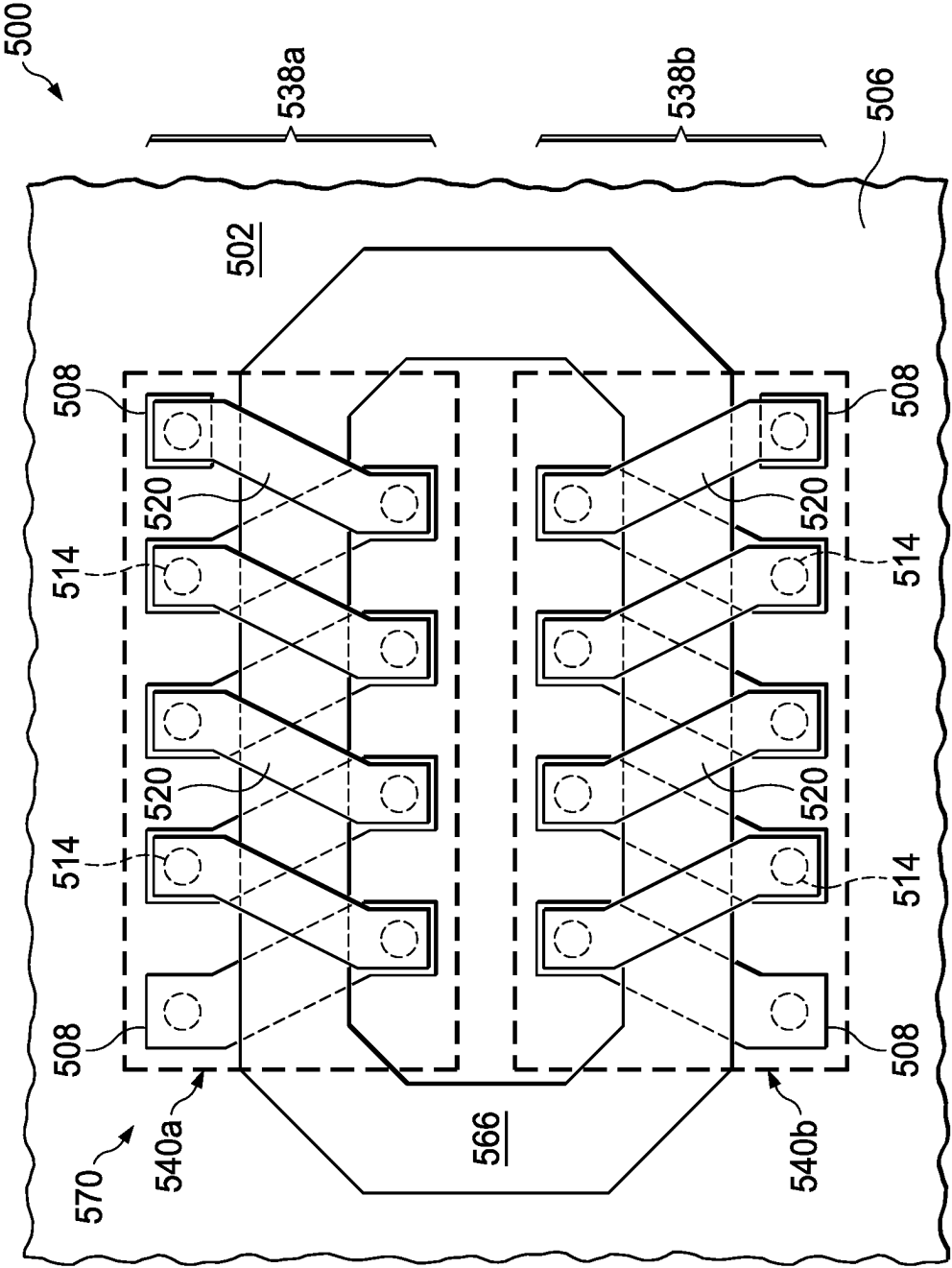


FIG. 5

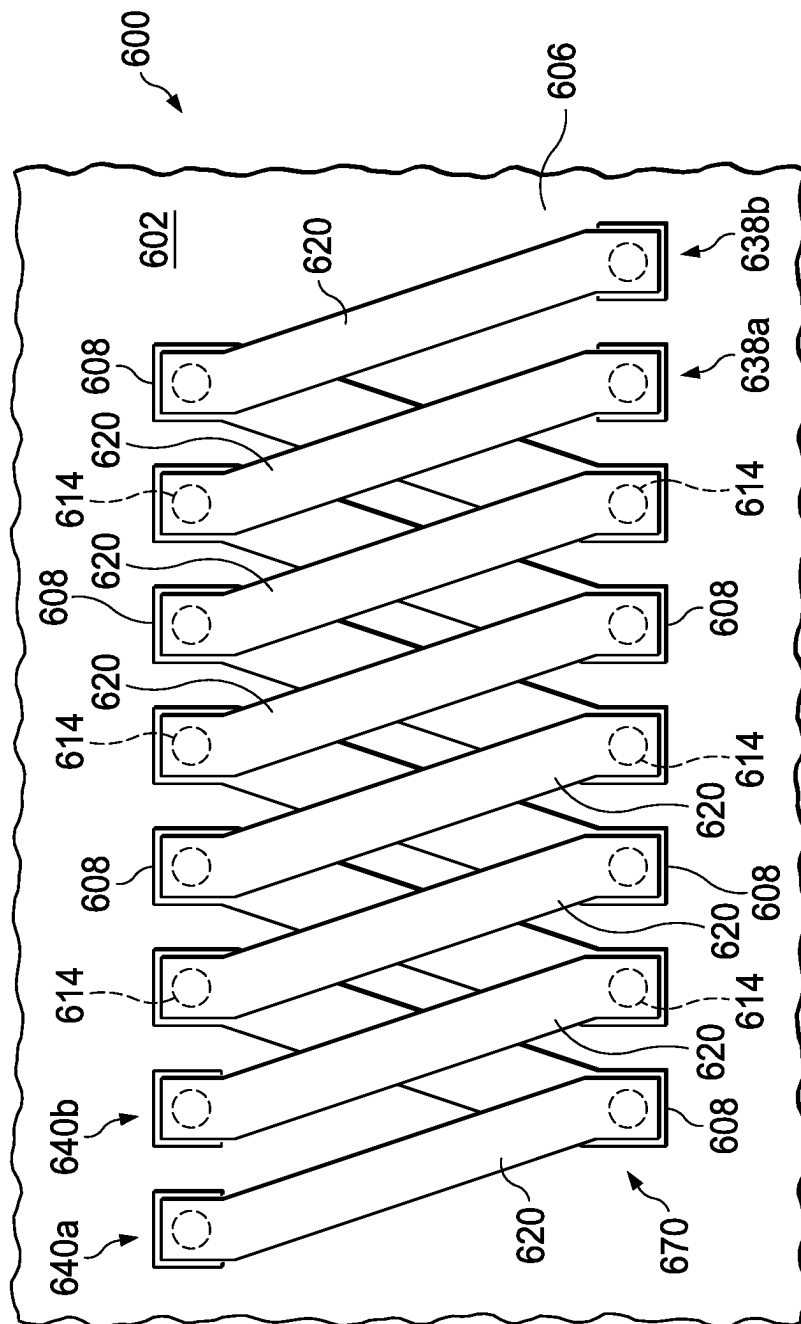


FIG. 6

INDUCTOR ON MICROELECTRONIC DIE

This application is a division of U.S. patent application Ser. No. 16/182,111, filed Nov. 6, 2018, the contents of which are herein incorporated by reference in its entirety.

FIELD

This disclosure relates to the field of microelectronic devices. More particularly, this disclosure relates to inductors on microelectronic devices.

BACKGROUND

Microelectronic devices frequently include circuits with inductors, commonly in the range of 0.5 to 5 nanohenries (nH). Inductors having in this range tend to require several hundred square microns, or more, undesirably increasing size and cost of the microelectronic devices. Attaining desirable Q values in these inductors has been challenging.

SUMMARY

The present disclosure introduces a microelectronic device having a die, and bump bonds and an inductor, on the die. The die contains terminals extending to a terminal surface of the die. The microelectronic device includes first lateral conductors extending along the terminal surface, wherein at least a portion of the first lateral conductors are in contact with at least a portion of the terminals. The microelectronic device also includes conductive columns on the first lateral conductors, extending perpendicularly away from the terminal surface, and second lateral conductors on the conductive columns, opposite from the first lateral conductors, extending laterally in a plane parallel to the terminal surface. The second lateral conductors have die attach surfaces located opposite from the conductive columns. A first set of the first lateral conductors, the conductive columns, and the second lateral conductors provide the bump bonds of the microelectronic device. A second set of the first lateral conductors, the conductive columns, and the second lateral conductors are electrically coupled in series to form the inductor. Methods of forming the microelectronic device are also disclosed.

BRIEF DESCRIPTION OF THE VIEWS OF THE DRAWINGS

FIG. 1 is a cross section of an example microelectronic device having a die, and bump bonds and an inductor on the die.

FIG. 2A through FIG. 2L are cross sections of a microelectronic device having a die, and bump bonds and an inductor on the die, depicted in stages of an example method of formation.

FIG. 3A through FIG. 3F are cross sections of a microelectronic device having a die, and bump bonds and an inductor on the die, depicted in stages of another example method of formation.

FIG. 4 is a top view of an example microelectronic device having a die, and an inductor on the die.

FIG. 5 is a top view of another example microelectronic device having a die, and a transformer including two inductors on the die.

FIG. 6 is a top view of another example microelectronic device having a die, and a transformer including two inductors on the die.

DETAILED DESCRIPTION

The present disclosure is described with reference to the attached figures. The figures are not drawn to scale and they are provided merely to illustrate the disclosure. Several aspects of the disclosure are described below with reference to example applications for illustration. It should be understood that numerous specific details, relationships, and methods are set forth to provide an understanding of the disclosure. The present disclosure is not limited by the illustrated ordering of acts or events, as some acts may occur in different orders and/or concurrently with other acts or events. Furthermore, not all illustrated acts or events are required to implement a methodology in accordance with the present disclosure. In addition, although some of the embodiments illustrated herein are shown in two dimensional views with various regions having depth and width, it should be clearly understood that these regions are illustrations of only a portion of a device that is actually a three dimensional structure. Accordingly, these regions will have three dimensions, including length, width, and depth, when fabricated on an actual device. Moreover, while the present invention is illustrated by embodiments directed to active devices, it is not intended that these illustrations be a limitation on the scope or applicability of the present invention. It is not intended that the active devices of the present invention be limited to the physical structures illustrated. These structures are included to demonstrate the utility and application of the present invention to disclosed embodiments.

A microelectronic device includes a die, and has bump bonds and an inductor on the die. The die contains terminals, for example, bond pads, extending to a terminal surface of the die. The terminal surface is not necessarily planar. The microelectronic device includes first lateral conductors extending along the terminal surface, external to the die. At least a portion of the first lateral conductors are in contact with at least a portion of the terminals.

The microelectronic device includes conductive columns on the first lateral conductors, extending perpendicularly from the terminal surface. At least a portion of the conductive columns may contact the corresponding first lateral conductors at locations which are laterally displaced from the terminals to which the corresponding first lateral conductors are contacting. That is, at least a portion of the conductive columns are not located directly over the terminals to which they are electrically coupled through the corresponding first lateral conductors.

The microelectronic device includes second lateral conductors on the conductive columns, opposite from the first lateral conductors, extending laterally in a plane parallel to the terminal surface. The second lateral conductors have die attach surfaces located opposite from the conductive columns. Solder or electrically conductive adhesive may be disposed on the die attach surfaces of at least a portion of the second lateral conductors.

A first set of the first lateral conductors, the conductive columns, and the second lateral conductors provide the bump bonds of the microelectronic device. A second set of the first lateral conductors, the conductive columns, and the second lateral conductors are electrically coupled in series to form the inductor. The inductor may have a linear configuration, a toroidal configuration, or other configuration. One or more nodes of the inductor may contact terminals at the terminal surface. One or more nodes of the inductor may

extend to the second lateral conductors for electrical connection to external leads of a package. The inductor may be a part of a transformer.

For the purposes of this disclosure, the term “lateral” is understood to refer to a direction parallel to a plane of the terminal surface of the die. It is noted that terms such as over and under may be used in this disclosure. These terms should not be construed as limiting the position or orientation of a structure or element, but should be used to provide spatial relationship between structures or elements. For the purposes of this disclosure, it will be understood that, if an element is referred to as being “connected” or “coupled” to another element, it may be directly connected or directly coupled to the other element, or intervening elements may be present.

FIG. 1 is a cross section of an example microelectronic device having a die, and bump bonds and an inductor on the die. The microelectronic device 100 includes the die 102, which may be manifested, for example, as a discrete semiconductor device, an integrated circuit, or a microelectrical mechanical system (MEMS) device. The die 102 has terminals 104 which extend to a terminal surface 106 of the die 102. The terminals 104 may include primarily aluminum or copper, for example. The terminals 104 may be manifested as bond pads, above a top interconnect level of the die 102, or may be portions of the top interconnect level. The terminals 104 may include an under bump metal (UBM) layer at the terminal surface 106. The UBM layer may include metals such as nickel, palladium, platinum, gold, copper, titanium, tungsten, chromium, or such. The die 102 may have a protective overcoat (PO) layer of electrically insulating material, not shown in FIG. 1, extending to the terminal surface 106 between the terminals 104. The PO layer may include silicon dioxide, silicon nitride, silicon oxynitride, polyimide, or such.

The microelectronic device 100 includes first lateral conductors 108 which extend along the terminal surface 106. At least a portion, and optionally all, of the first lateral conductors 108 contact at least a portion, and optionally all, of the terminals 104. Individual first lateral conductors 108 may contact one or more of the terminals 104. In this example, the first lateral conductors 108 are manifested with first conductor seed layers 110 on the terminal surface 106, and first main conductors 112 on the first conductor seed layers 110. The first main conductors 112 extend laterally to lateral boundaries of the first lateral conductors 108. The first main conductors 112 may extend in a straight line, or may have one or more lateral bends. The first conductor seed layers 110 may include adhesion layers containing metals such as titanium, tungsten, chromium, or nickel, on the terminal surface 106, and plating layers of copper on the adhesion layers. The first conductor seed layers 110 may have thicknesses of 10 nanometers to 1 micron, by way of example. The first main conductors 112 are electrically conductive and may include plated copper, optionally with other metals such as gold, silver, or nickel. The first main conductors 112 may have thicknesses of 3 microns to 30 microns, by way of example.

The microelectronic device 100 includes conductive columns 114 disposed on the first lateral conductors 108. The conductive columns 114 extend from the first lateral conductors 108, perpendicularly to the terminal surface 106. The term “perpendicularly” is understood to encompass orientations that are substantially perpendicular, within fabrication tolerances encountered when forming the microelectronic device 100. The term “perpendicularly” is further understood to encompass orientations that are substantially

perpendicular, within measurement tolerances encountered when measuring the microelectronic device 100. A portion of the conductive columns 114 may contact the corresponding first lateral conductors 108 at locations which are laterally displaced from the terminals 104 to which the corresponding first lateral conductors 108 are contacting. Another portion of the conductive columns 114 may contact the corresponding first lateral conductors 108 directly over the terminals 104 to which the corresponding first lateral conductors 108 are contacting. Instances of both portions of the conductive columns 114 are depicted in FIG. 1. In this example, the conductive columns 114 are manifested with column seed layers 116 on the first lateral conductors 108 and main columns 118 on the column seed layers 116. The main columns 118 are electrically conductive and extend to tops of the conductive columns 114, opposite from the first lateral conductors 108. The column seed layers 116 may include adhesion layers containing metals such as titanium, chromium, or nickel, on the terminal surface 106, and plating layers of copper on the adhesion layers. The column seed layers 116 may have thicknesses of 10 nanometers to 1 micron, by way of example. The main columns 118 may include plated copper, optionally with other metals such as gold, silver, or nickel, and may have compositions similar to the first main conductors 112. The main columns 118 may have heights, measured perpendicular to the terminal surface 106, of 30 microns to 100 microns, by way of example. The conductive columns 114 may have a variety of cross-sectional shapes. Some instances of the conductive columns 114 may have circular cross-sectional shapes, or square cross-sectional shapes with rounded corners. Other instances of the conductive columns 114 may have oval or rectangular cross-sectional shapes. Other cross-sectional shapes for the conductive columns 114 are within the scope of this example. The main columns 118 may have widths, measured parallel to the terminal surface 106, of 25 microns to 50 microns, and may have lengths, also measured parallel to the terminal surface 106, of 25 microns to 300 microns, by way of example.

The microelectronic device 100 further includes second lateral conductors 120 disposed on the conductive columns 114. The second lateral conductors 120 and the first lateral conductors 108 are located at opposite ends of the conductive columns 114. At least a portion of the second lateral conductors 120 extend laterally past the corresponding conductive columns 114 on which they are disposed, in a plane parallel to the terminal surface. In this example, the second lateral conductors 120 are manifested with second conductor seed layers 122 on the conductive columns 114, and second main conductors 124 on the second conductor seed layers 122. The second main conductors 124 extend laterally to lateral boundaries of the second lateral conductors 120. The second main conductors 124 may extend in a straight line, or may have one or more lateral bends. The second conductor seed layers 122 may include adhesion layers containing metals such as titanium, chromium or nickel, on the conductive columns 114, and plating layers of copper on the adhesion layers. The second conductor seed layers 122 may have thicknesses of 10 nanometers to 1 micron, by way of example. The second main conductors 124 may include plated copper, optionally with other metals such as gold, silver, or nickel, and may have compositions similar to the first main conductors 112. The second main conductors 124 may have thicknesses of 3 microns to 30 microns, by way of example. The second lateral conductors 120 have die attach surfaces 126 located opposite from the first lateral conductors 108.

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A first set **128** of the first lateral conductors **108**, the conductive columns **114**, and the second lateral conductors **120** provide the bump bonds **130** of the microelectronic device **100**. A die attach material **132** is disposed on the die attach surfaces **126** of the bump bonds **130**. The die attach material **132** may include a solder, for example, in the form of solder paste, or a solder layer formed using a melted solder bath. Alternatively, the die attach material **132** may include an electrically conductive adhesive, such as epoxy with metal microparticles. Other compositions for the die attach material **132** are within the scope of this example. One or more of the bump bonds **130** may include an insulating layer **134** on the die attach surfaces **126**, to define areas for the die attach material **132**. The insulating layer **134** may include polymer insulating material such as polyimide or polyester, or may include inorganic insulating material such as ceramic or glass frits.

FIG. **1** depicts the microelectronic device **100** attached to external leads **136**. The external leads **136** may be part of a package containing the microelectronic device **100**, or may be part of a carrier or circuit board on which the microelectronic device **100** is mounted. The external leads **136** are electrically coupled to the bump bonds **130** through the die attach material **132**. Having the first lateral conductors **108** and the second lateral conductors **120** in the bump bonds **130** may enable having a desired arrangement of the terminals **104** on the die **102** and connecting the terminals **104** to a desired arrangement of the external leads **136**. Encapsulation material, such as epoxy, not shown in FIG. **1**, may be disposed on the terminal surface **106**, covering the first lateral conductors **108**, and surrounding the conductive columns **114**.

A second set **138** of the first lateral conductors **108**, the conductive columns **114**, and the second lateral conductors **120** are electrically configured in series to provide the inductor **140**. The inductor **140** may have a linear configuration, as depicted in FIG. **1**, in which the first lateral conductors **108**, the conductive columns **114**, and the second lateral conductors **120** of the inductor **140** are arranged in a linear array. Alternatively, the inductor **140** may have a toroidal configuration, or other configuration. The inductor **140** may have a desirably high quality factor, commonly referred to as the Q factor, due to low electrical resistance in the first lateral conductors **108**, the conductive columns **114**, and the second lateral conductors **120**. Having the first lateral conductors **108** including copper with thicknesses of 3 microns to 30 microns may provide a Q factor greater than 1 at a frequency of 100 megahertz (MHz), which may be difficult to attain using a thinner redistribution layer (RDL) or interconnects in the die **102**. The low electrical resistance of the inductor **140** may enable use in power circuits as well as signal circuits, which may degrade reliability in an inductor formed with a thinner RDL layer. The inductor **140** may include additional electrically conductive elements, in parallel to the first lateral conductors **108** or the second lateral conductors **120**, to further improve the Q factor. For example, interconnects of the die **102** may be electrically coupled in parallel with the first lateral conductors **108** in the second set **138**. Instances of the external leads **136** may be electrically coupled in parallel with the second lateral conductors **120** in the second set **138**. One or more nodes of the inductor **140** may contact one or more of the terminals **104**, as depicted in FIG. **1**. One or more nodes of the inductor **140** may extend to the die attach surfaces **126** for electrical connection to external leads of a package. The inductor **140** may be a part of a transformer.

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FIG. **2A** through FIG. **2L** are cross sections of a microelectronic device having a die, and bump bonds and an inductor on the die, depicted in stages of an example method of formation. Referring to FIG. **2A**, the microelectronic device **200** includes the die **202**, which may be implemented as a discrete semiconductor device, an integrated circuit, a MEMS device, or other such microelectronic die. The die **202** may be part of a semiconductor wafer containing additional die. The die **202** has terminals **204** which extend to a terminal surface **206** of the die **202**. The terminals **204** may be formed of primarily aluminum or copper, for example, and may have a UBM layer formed at the terminal surface **206** to protect the terminals **204** during fabrication, and so provide a low electrical connection to the terminals **204**. The UBM layer may be formed, for example, by a sputter process or an electroless plating process. The die **202** may have a PO layer extending to the terminal surface **206** between the terminals **204**. The PO layer may include any of the electrically insulating materials disclosed in reference to the PO layer of FIG. **1**, and may be formed, for example, by a plasma enhanced chemical vapor deposition (PECVD) process or a photolithographic process.

A first conductor seed layer **210** is formed on the terminal surface **206**, contacting the terminals **204**. The first conductor seed layer **210** may include an adhesion layer formed on the terminal surface **206**, contacting the terminals **204**, and a plating layer on the adhesion layer. The adhesion layer may include one or more metals having desired adhesion to the terminals **204** and to material of the die **202**, such as the PO layer, at the terminal surface **206** adjacent to the terminals **204**. For example, the adhesion layer may include titanium, titanium tungsten, chromium, or nickel, and may be formed by one or more sputter processes. The plating layer may include primarily copper, and may be formed by a sputter process.

A first conductor plating mask **242** is formed over the first conductor seed layer **210**, exposing the first conductor seed layer **210** in areas for subsequently-formed first lateral conductors **208**, shown in FIG. **2B**. In one version of this example, the first conductor plating mask **242** may include photoresist, or other photosensitive polymer, and may be formed by a photolithographic process. In another version, the first conductor plating mask **242** may include polymer material, and may be formed by an additive process, such as a material jetting process. In a further version, the first conductor plating mask **242** may include thermally erodible material such as polyimide, polyester, or polymethylmethacrylate (PMMA), and may be formed by a laser ablation process. Other materials for the first conductor plating mask **242** and methods for forming the first conductor plating mask **242** are within the scope of this example.

Referring to FIG. **2B**, the first main conductors **212** are formed on the first conductor seed layer **210** where exposed by the first conductor plating mask **242**. The first main conductors **212** may have the composition disclosed in reference to the first main conductors **112** of FIG. **1**. The first main conductors **212** may be formed by a copper electroplating process, or optionally by a copper electroless plating operation. The first conductor seed layer **210** and the first main conductors **212** provide the first lateral conductors **208** of the microelectronic device **200**.

Referring to FIG. **2C**, a column seed layer **216** is formed on the first lateral conductors **208** and over the first conductor plating mask **242**. The column seed layer **216** may include an adhesion layer formed on the first lateral conductors **208** and over the first conductor plating mask **242**, contacting the first lateral conductors **208**, and a plating

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layer on the adhesion layer. The adhesion layer may include one or more metals having desired adhesion to metal in the first lateral conductors **208** and to material of the first conductor plating mask **242**. For example, the adhesion layer may include titanium, chromium, or nickel, and may be formed by one or more sputter processes. The plating layer may include primarily copper, and may be formed by a sputter process.

Referring to FIG. 2D, a column plating mask **244** is formed over the column seed layer **216**, exposing the column seed layer **216** in areas for subsequently-formed conductive columns **214**, shown in FIG. 2E. The column plating mask **244** may be formed by any of the methods disclosed in reference to the first conductor plating mask **242**. Other materials for the column plating mask **244** and methods for forming the column plating mask **244** are within the scope of this example.

Referring to FIG. 2E, main columns **218** are formed on the column seed layer **216** where exposed by the column plating mask **244**. The main columns **218** may have the composition disclosed in reference to the main columns **118** of FIG. 1. The main columns **218** may be formed by a copper electroplating process, or optionally by a copper electroless plating operation. The column seed layer **216** and the main columns **218** provide the conductive columns **214** of the microelectronic device **200**.

Referring to FIG. 2F, a second conductor seed layer **222** is formed on the conductive columns **214** and over the column plating mask **244**. The second conductor seed layer **222** may include an adhesion layer formed on the conductive columns **214** and over the column plating mask **244**, contacting the conductive columns **214**, and a plating layer on the adhesion layer. The adhesion layer may include one or more metals having desired adhesion to metal in the conductive columns **214** and to material of the column plating mask **244**. For example, the adhesion layer may include any of the metals disclosed in reference to the adhesion layer of the column seed layer **216**, and may be formed by one or more sputter processes. The plating layer may include primarily copper, and may be formed by a sputter process.

A second conductor plating mask **246** is formed over the second conductor seed layer **222**, exposing the second conductor seed layer **222** in areas for subsequently-formed second lateral conductors **220**, shown in FIG. 2G. The second conductor plating mask **246** may be formed by any of the methods disclosed in reference to the first conductor plating mask **242**. Other materials for the second conductor plating mask **246** and methods for forming the second conductor plating mask **246** are within the scope of this example.

Referring to FIG. 2G, the second main conductors **224** are formed on the second conductor seed layer **222** where exposed by the second conductor plating mask **246**. The second main conductors **224** may have the composition disclosed in reference to the second main conductors **124** of FIG. 1. The second main conductors **224** may be formed by a copper electroplating process, or optionally by a copper electroless plating operation. The second conductor seed layer **222** and the second main conductors **224** provide the second lateral conductors **220** of the microelectronic device **200**.

Referring to FIG. 2H, the second conductor plating mask **246** of FIG. 2G is removed, leaving the second lateral conductors **220** in place. The second conductor plating mask **246** may be removed by a dry process using oxygen radicals, such as an ash process or an ozone process. Alternatively, the second conductor plating mask **246** may be removed by

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a wet process using solvents such as n-methyl-2-pyrrolidone (NMP) or dimethyl sulfoxide (DMSO). Proprietary formulations of resist removal chemicals for removing the second conductor plating mask **246** are commercially available from several suppliers.

Referring to FIG. 2I, the second conductor seed layer **222** is removed where exposed by the second main conductors **224**, leaving the second conductor seed layer **222** in place under the second main conductors **224**. The second conductor seed layer **222** may be removed by a wet etch process using an acid bath. A small portion of the second main conductors **224** may be removed during removal of the second conductor seed layer **222**.

Referring to FIG. 2J, the column plating mask **244** of FIG. 2I is removed, leaving the main columns **218** in place. The column plating mask **244** may be removed by a process similar to the process used to remove the second conductor plating mask **246** of FIG. 2G. Other methods for removing the column plating mask **244** are within the scope of this example.

The column seed layer **216** is removed where exposed by the main columns **218**. The column seed layer **216** may be removed by a process similar to the process used to remove the second conductor seed layer **222**. Other methods to remove the column seed layer **216** are within the scope of this example. Removal of the column seed layer **216** may result in removal of the second conductor seed layer **222** on the second main conductors **224** where exposed by the main columns **218**, as depicted in FIG. 2J.

Subsequently, the first conductor plating mask **242** is removed, leaving the first main conductors **212** in place. The first conductor plating mask **242** may be removed by a process similar to the process used to remove the second conductor plating mask **246**. Other methods for removing the first conductor plating mask **242** are within the scope of this example.

The first conductor seed layer **210** is removed where exposed by the first main conductors **212**, leaving the first conductor seed layer **210** in place between the first main conductors **212** and the terminal surface **206**. The first conductor seed layer **210** may be removed by a process similar to the process used to remove the second conductor seed layer **222**. Other methods to remove the first conductor seed layer **210** are within the scope of this example.

The second lateral conductors **220** have die attach surfaces **226** located opposite from the first lateral conductors **208**. A first set **228** of the first lateral conductors **208**, the conductive columns **214**, and the second lateral conductors **220** provide the bump bonds **230** of the microelectronic device **200**. A second set **238** of the first lateral conductors **208**, the conductive columns **214**, and the second lateral conductors **220** are electrically configured in series to provide the inductor **240**. Forming the first lateral conductors **208**, the conductive columns **214**, and the second lateral conductors **220**, of the first set **228** and the second set **238**, concurrently, may advantageously reduce fabrication cost and complexity compared to forming the inductor **240** separately from the bump bonds **230**.

Referring to FIG. 2K, encapsulation material **248** may be formed on the die **202**, surrounding the first lateral conductors **208** and the conductive columns **214**, and extending to the second lateral conductors **220**. The encapsulation material **248** may include epoxy, and may be formed by injection molding or press molding, by way of example. The encapsulation material **248** may include magnetic particles **250**, such as ferrite particles or ferromagnetic particles containing iron, nickel, or cobalt. The magnetic particles **250** may

provide an average relative magnetic permeability of the encapsulation material **248** greater than 1, wherein the relative magnetic permeability of a vacuum is 1, thereby advantageously increasing an inductance of the inductor **240**.

Referring to FIG. 2L, an insulating layer **234** may be formed on the second main conductors **224**, to define areas for a subsequently-disposed die attach material **232**, and to isolate the second main conductors **224** in the second set **238**. The insulating layer **234** may include any of the materials disclosed in reference to the insulating layer **134** of FIG. 1. The insulating layer **234** may be formed by any of several methods. In one version of this example, the insulating layer **234** may be formed by spin coating the microelectronic device **200** with a photosensitive polymer material such as polyimide, and exposing the photosensitive polymer material to patterned ultraviolet (UV) light, then developing the photosensitive polymer material. In another version, the insulating layer **234** may be formed by a screen printing process. In a further version, the insulating layer **234** may be formed by an additive process, such as a material extrusion process. Other methods for forming the insulating layer **234** are within the scope of this example.

The die attach material **232** is formed on the die attach surfaces **226** of the bump bonds **230**. The die attach material **232** may include solder, in the form of solder paste, formed by a screen print process or a material extrusion process. The die attach material **232** may include solder, in the form of a solder layer, formed using a melted solder bath. The die attach material **232** may include electrically conductive adhesive, formed by a screen print process or a material extrusion process. Other compositions for the die attach material **232** and methods for formation are within the scope of this example. The insulating layer **234** may be used to define areas for the die attach material **232**.

The microelectronic device **200** is attached to external leads **236** by electrically coupling the bump bonds **230** to the external leads **236** through the die attach material **232**. The external leads **236** may be part of a package, such as a lead frame or chip carrier, containing the microelectronic device **200**. Alternatively, the external leads **236** may be part or a circuit substrate, such as a printed circuit board (PCB), on which the microelectronic device **200** is mounted. In versions of this example in which the die attach material **232** includes solder, the microelectronic device **200** may be attached to the external leads **236** by a solder reflow process. In versions of this example in which the die attach material **232** includes adhesive, the microelectronic device **200** may be attached to the external leads **236** by an adhesive curing process. The inductor **240**, being formed of the first lateral conductors **208**, the conductive columns **214**, and the second lateral conductors **220**, may advantageously be sufficiently robust to undergo the process of attaching the microelectronic device **200** to the external leads **236**, without significant degradation.

FIG. 3A through FIG. 3F are cross sections of a microelectronic device having a die, and bump bonds and an inductor on the die, depicted in stages of another example method of formation. Referring to FIG. 3A, the microelectronic device **300** includes the die **302**, which may be implemented as a discrete semiconductor device, an integrated circuit, a MEMS device, or other such microelectronic die. The die **302** has terminals **304** of electrically conductive material, which extend to a terminal surface **306** of the die **302**.

In this example, the terminals **304** may include one or more elongated terminals **304a**, spanning a length for a

lower winding in an area for the inductor **340**. One or more interconnects **352** of the die **302** may also span the length for the lower winding, and may be electrically coupled to the elongated terminal **304a** by vias **354** of the die **302**. The interconnects **352** and the vias **354** may be parts of an interconnect network of the die **302**.

A first conductor seed layer **310** is formed on the die **302**, contacting the terminals **304**. The first conductor seed layer **310** may have a layer structure and composition as disclosed in reference to the first conductor seed layer **210** of FIG. 2A, and may be formed as disclosed in reference to the first conductor seed layer **210**.

A first conductor plating mask **342** is formed over the first conductor seed layer **310**, exposing areas for first lateral conductors **308**. The first conductor plating mask **342** may have a composition as disclosed in reference to the first conductor plating mask **242** of FIG. 2A, and may be formed as disclosed in reference to the first conductor plating mask **242**.

First main conductors **312** are formed on the first conductor seed layer **310**, where exposed by the first conductor plating mask **342**, using a first copper plating bath **356**. The first copper plating bath **356** may be implemented in an electroplating process, or in an electroless plating process. In this example, the first conductor plating mask **342** is left in place after the first main conductors **312** are formed. A portion of the first conductor seed layer **310** that is between the first main conductors **312** and the terminal surface **306**, combined with the first main conductors **312**, provide the first lateral conductors **308** of the microelectronic device **300**.

Referring to FIG. 3B, a column plating mask **344** is formed over the first conductor plating mask **342** and the first lateral conductors **308**, exposing areas for conductive columns **314** on the first lateral conductors **308**. The column plating mask **344** may have a composition as disclosed for the column plating mask **244** of FIG. 2D, and may be formed by any of the methods disclosed for the column plating mask **244**.

A second conductor seed layer **358** is formed over the column plating mask **344**, making contact with the first lateral conductors **308** where exposed by the column plating mask **344**. The second conductor seed layer **358** may have a layer structure and composition as disclosed in reference to the column seed layer **216** of FIG. 2C or the second conductor seed layer **222** of FIG. 2F, and may be formed as disclosed in reference to the column seed layer **216** or the first conductor seed layer **210**.

A second conductor plating mask **346** is formed over the second conductor seed layer **358**, exposing areas for second lateral conductors **320**. The second conductor plating mask **346** may have a composition as disclosed for the second conductor plating mask **246** of FIG. 2F, and may be formed by any of the methods disclosed for the second conductor plating mask **246**.

Second main conductors **360** are formed on the second conductor seed layer **358**, where exposed by the second conductor plating mask **346**, using a second copper plating bath **362**. The second copper plating bath **362** may be implemented in an electroplating process, or in an electroless plating process, and may be implemented using equipment and plating solutions of the first copper plating bath **356** of FIG. 3A. A portion of the second conductor seed layer **358** that is laterally surrounded by the column plating mask **344**, combined with a portion of the second main conductors **360** that are laterally surrounded by the column plating mask **344**, provide the conductive columns **314** of the microelec-

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tronic device 300. A portion of the second conductor seed layer 358 that is laterally surrounded by the second conductor plating mask 346, combined with a portion of the second main conductors 360 that are laterally surrounded by the second conductor plating mask 346, provide the second lateral conductors 320 of the microelectronic device 300. Providing the conductive columns 314 and the second lateral conductors 320 from portions of the second main conductors 360, which was formed using one plating bath, may advantageously reduce fabrication cost and complexity compared to forming the conductive columns 314 and the second lateral conductors 320 using separate plating baths.

Referring to FIG. 3C, the second lateral conductors 320 have die attach surfaces 326 located opposite from the first lateral conductors 308. Barrier layers 364 may optionally be formed on the second lateral conductors 320, covering the die attach surfaces 326. The barrier layers 364 may include one or more metals which reduce diffusion of copper and tin to inhibit formation of copper-tin intermetallic compounds. The barrier layers 364 may include, for example, nickel, cobalt, or molybdenum. The barrier layers 364 may be particularly advantageous when solder containing tin, such as silver-tin solder, is disposed on the second lateral conductors 320. The barrier layers 364 may be formed by an electroplating process, such as a reverse pulse electroplating process, which may enable a desired ratio of the metals in the barrier layers 364 that would be difficult to attain using direct current (DC) plating.

The second conductor plating mask 346 is subsequently removed. The second conductor plating mask 346 may be removed by any of the methods disclosed for removing the second conductor plating mask 246 in reference to FIG. 2H.

The second conductor seed layer 358 is removed where exposed by the removal of the second conductor plating mask 346. The second conductor seed layer 358 may be removed by any of the methods disclosed for removing the second conductor seed layer 222 in reference to FIG. 2I.

The column plating mask 344 removed. The column plating mask 344 may be removed by any of the methods disclosed for removing the column plating mask 244 in reference to FIG. 2J.

The first conductor seed layer 310 is removed where exposed by the removal of the column plating mask 344. The first conductor seed layer 310 may be removed by any of the methods disclosed for removing the first conductor seed layer 210 in reference to FIG. 2J. Removal of the first conductor seed layer 310 may result in removal of portions of the second conductor seed layer 358 that are exposed by the removal of the column plating mask 344.

Referring to FIG. 3D, a first set 328 of the first lateral conductors 308, the conductive columns 314, and the second lateral conductors 320 provide the bump bonds 330 of the microelectronic device 300. A second set 338 of the first lateral conductors 308, the conductive columns 314, and the second lateral conductors 320 are electrically configured in series to provide the inductor 340. The elongated terminal 304a and the interconnect 352 are electrically coupled in parallel to the first lateral conductors 308 of the inductor 340, reducing an electrical resistance of the inductor 340 and thus advantageously increasing a Q factor of the inductor 340.

A die attach material 332 is formed over the die attach surfaces 326, on the barrier layers 364, if present. The die attach material 332 may have any of the compositions disclosed for the die attach material 232 of FIG. 2L. The die attach material 332 may be formed by any of the methods disclosed in reference to the die attach material 232.

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A magnetic material 366 having a relative magnetic permeability greater than 1 may be formed between the conductive columns 314 of the inductor 340, which may advantageously increase an inductance of the inductor 340. The magnetic material 366 may include, for example, ferrite particles or ferromagnetic particles containing iron, nickel, or cobalt, in a polymer binder such as epoxy. The magnetic material 366 may be formed in the inductor 340 using an additive process, such as a material extrusion process 368 as depicted in FIG. 3D.

Referring to FIG. 3E, the microelectronic device 300 is attached to external leads 336 by electrically coupling the bump bonds 330 and the inductor 340 to the external leads 336 through the die attach material 332. The external leads 336 may be part of a package containing the microelectronic device 300, or may be part of a circuit substrate on which the microelectronic device 300 is mounted. The microelectronic device 300 may be attached to the external leads 336 as described in reference to FIG. 2L. In this example, the external leads 336 that are electrically coupled to the inductor 340 may further reduce the electrical resistance of the inductor 340 and thus advantageously increase the Q factor of the inductor 340. The inductor 340 and the bump bonds 330, being formed of the first lateral conductors 308, the conductive columns 314, and the second lateral conductors 320, may advantageously be sufficiently robust to undergo the process of attaching the microelectronic device 300 to the external leads 336, without loss of mechanical integrity, even though the inductor 340 and the bump bonds 330 are not mechanically supported by encapsulation material.

Referring to FIG. 3F, encapsulation material 348, which may be referred to as an underfill material, may be formed on the die 302, surrounding the first lateral conductors 308, the conductive columns 314, the second lateral conductors 320, and the die attach material 332, and extending to the external leads 336. The encapsulation material 348 may include epoxy, and may be formed by injection molding. The encapsulation material 348 may provide mechanical support for the first lateral conductors 308, the conductive columns 314, and the second lateral conductors 320.

FIG. 4 is a top view of an example microelectronic device having a die, and an inductor on the die. The microelectronic device 400 includes the die 402, and the die 402 has a terminal surface 406. The microelectronic device 400 includes first lateral conductors 408, conductive columns 414, and second lateral conductors 420 on the terminal surface 406. A first set, not shown in FIG. 4, of the first lateral conductors 408, the conductive columns 414, and the second lateral conductors 420 provide bump bonds, not shown in FIG. 4, of the microelectronic device 400. A second set 438 of the first lateral conductors 408, the conductive columns 414, and the second lateral conductors 420 are electrically configured in series to provide the inductor 440. In this example, the inductor 440 has a toroidal configuration, that is, the first lateral conductors 408, the conductive columns 414, and the second lateral conductors 420 of the inductor 440 are arranged on a closed loop array. A magnetic material 466 having a relative magnetic permeability greater than 1 may be located in the inductor 440, over the first lateral conductors 408 and under the second lateral conductors 420. The toroidal configuration may provide a desired inductance for the inductor 440 in a compact space on the terminal surface 406. One or more nodes of the inductor 440 may be electrically coupled to components in the die 402, or to external leads, not shown in FIG. 4.

FIG. 5 is a top view of another example microelectronic device having a die, and a transformer including two induc-

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tors on the die. The microelectronic device **500** includes the die **502**, and the die **502** has a terminal surface **506**. The microelectronic device **500** includes first lateral conductors **508**, conductive columns **514**, and second lateral conductors **520** on the terminal surface **506**. A first set, not shown in FIG. **5**, of the first lateral conductors **508**, the conductive columns **514**, and the second lateral conductors **520** provides bump bonds, not shown in FIG. **5**, of the microelectronic device **500**. A first second set **538a** of the first lateral conductors **508**, the conductive columns **514**, and the second lateral conductors **520** are electrically configured in series to provide a first inductor **540a**. A second set **538b** of the first lateral conductors **508**, the conductive columns **514**, and the second lateral conductors **520** are electrically configured in series to provide a second inductor **540b**. A magnetic material **566** having a relative magnetic permeability greater than 1 is located in the first inductor **540a** and in the second inductor **540b**, over the first lateral conductors **508** and under the second lateral conductors **520**. In this example, the first inductor **540a** has a linear configuration, that is, the first lateral conductors **508**, the conductive columns **514**, and the second lateral conductors **520** are arranged on a surface of a cylinder around the magnetic material **566**. Similarly, the second inductor **540b** has a linear configuration around the magnetic material **566**. The first inductor **540a** and the second inductor **540b** are elements of the transformer **570**. FIG. **5** depicts the first inductor **540a** and the second inductor **540b** with equal numbers of windings around the magnetic material **566**, however, other configurations of the transformer **570** having unequal numbers of windings are within the scope of this example. The transformer **570** may enable transmission of signals or power between the first inductor **540a** and the second inductor **540b**, advantageously without consuming space in the die **502**.

FIG. **6** is a top view of another example microelectronic device having a die, and a transformer including two inductors on the die. The microelectronic device **600** includes the die **602**, and the die **602** has a terminal surface **606**. The microelectronic device **600** includes first lateral conductors **608**, conductive columns **614**, and second lateral conductors **620** on the terminal surface **606**. A first set, not shown in FIG. **6**, of the first lateral conductors **608**, the conductive columns **614**, and the second lateral conductors **620** provides bump bonds, not shown in FIG. **6**, of the microelectronic device **600**. A first second set **638a** of the first lateral conductors **608**, the conductive columns **614**, and the second lateral conductors **620** are electrically configured in series to provide a first inductor **640a**. A second set **638b** of the first lateral conductors **608**, the conductive columns **614**, and the second lateral conductors **620** are electrically configured in series to provide a second inductor **640b**. In this example, the first inductor **640a** and the second inductor **640b** have linear configurations, and are interdigitated to form the transformer **670**. The interdigitated configuration of the transformer **670** may enable transmission of signals or power between the first inductor **640a** and the second inductor **640b**, without magnetic material disposed in the first inductor **640a** or the second inductor **640b**.

Various features of the examples disclosed herein may be combined in other manifestations of example microelectronic devices. For example, the microelectronic device **100** of FIG. **1** may be formed by steps disclosed in reference to the method of FIG. **2A** through FIG. **2L**, by steps disclosed in reference to the method of FIG. **3A** through FIG. **3F**, or by another method. The microelectronic devices disclosed herein may be formed using any method, such as the methods described in the commonly assigned patent appli-

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cation having patent application Ser. No. 16/030,371, filed Jul. 9, 2018 which is incorporated herein by reference but is not admitted to be prior art with respect to the present invention by its mention in this section. The bump bonds **130** of FIG. **1** may have the barrier layers **364** of FIG. **3C**. The microelectronic device **100** of FIG. **1** may include the encapsulation material **248** with the magnetic particles **250**, as disclosed in reference to FIG. **2K**, or may include the magnetic material **366**, as disclosed in reference to FIG. **3D**.

While various embodiments of the present disclosure have been described above, it should be understood that they have been presented by way of example only and not limitation. Numerous changes to the disclosed embodiments can be made in accordance with the disclosure herein without departing from the spirit or scope of the disclosure. Thus, the breadth and scope of the present invention should not be limited by any of the above described embodiments. Rather, the scope of the disclosure should be defined in accordance with the following claims and their equivalents.

What is claimed is:

1. A method, comprising:

forming a first metal column on a first bond pad of a die;
forming a first metal strip on the die;
forming a second metal column on the first metal strip;
and

forming a second metal strip on the first and second metal columns and over the die, in which the second metal strip has a pair of bent segments and a straight segment coupled between the pair of bent segments.

2. The method of claim 1, further comprising:

forming a third metal column on the first metal strip or on the die;
forming a fourth metal column on a second bond pad of the die; and

forming a third metal strip on the third and fourth metal columns and over the die, in which the third metal strip has a second pair of bent segments and a second straight segment coupled between the second pair of bent segments.

3. The method of claim 2, further comprising:

forming a seed layer on the die, the seed layer including at least one of: titanium, tungsten, chromium, or nickel, wherein the first metal strip is formed on the seed layer.

4. The method of claim 1, wherein a thickness of the first metal strip is between 3 microns and 30 microns.

5. The method of claim 2, wherein:

each of the first through fourth metal columns includes copper;
a respective width of each of the first through third metal columns is between 25 microns and 50 microns;
a respective length of each of the first through third metal columns is between 25 microns and 300 microns; and
a respective height of each of the first through third metal columns is between 30 microns and 100 microns.

6. The method of claim 2, further comprising:

forming a first seed layer on the first metal strip; and
forming a second seed layer on the first, second, third, and fourth metal columns,

wherein:

the second metal column and the third metal column are formed on the first seed layer;
the second and third metal strips are formed on the second seed layer; and
each of the first and second seed layers includes at least one of: titanium, chromium, or nickel.

7. The method of claim 2, wherein

the second and third metal strips include copper.

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8. The method of claim 2, wherein a respective thickness of each of the second and third metal strips is between 3 microns to 30 microns.

9. The method of claim 2, further comprising:

forming a first layer of die attach material on the second metal strip; and

forming a second layer of die attach material on the third metal strip,

wherein the die attach material includes at least one of: a solder, or an adhesive.

10. The method of claim 2, further comprising:

forming a magnetic material around at least parts of the first through third metal strips and at least parts of the first through fourth metal columns.

11. The method of claim 10, wherein the magnetic material includes magnetic particles, and a molding compound in which the magnetic particles are suspended.

12. The method of claim 10, wherein the first through fourth metal columns and the first through third metal strips are part of an inductor electrically coupled between the first and second bond pads, in which the inductor has a linear configuration.

13. The method of claim 10, further comprising forming a fourth metal strip on the die and between the first and second bond pads,

wherein the first through fourth metal columns and the first through fourth metal strips are part of an inductor having a toroidal configuration.

14. The method of claim 2, further comprising:

forming a fifth metal column on a third bond pad on the die; and

forming a fourth metal strip on the fifth metal column over the die; and

forming a layer of die attach material on the fourth metal strip.

15. The method of claim 2, further comprising:

forming a fourth metal strip having a first end on a third bond pad on the die;

forming a fifth metal column on a second end of the fourth metal strip, the second end being opposite to the first end; and

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forming a layer of die attach material on the fifth metal column, in which the fifth metal column is between the layer of die attach material and the fourth metal strip.

16. The method of claim 6, further comprising:

forming a first patterned mask layer on the die, wherein the first metal strip is formed through the patterned first mask layer, and the first seed layer is formed on the patterned first mask layer and the first metal strip;

forming a second patterned mask layer on the first seed layer to expose part of the first seed layer, wherein the first, second, third, and fourth metal columns are formed through the patterned first mask layer and the patterned second mask layer on the exposed part of the first seed layer, and the second seed layer is formed on the first, second, third, and fourth metal columns and on the patterned second mask layer; and

forming a third patterned mask layer on the second seed layer to expose part of the second seed layer, wherein the second and third metal strips are formed through the patterned third mask layer on the exposed part of the second seed layer.

17. The method of claim 16, wherein the first, second, third, and fourth metal columns and the first through third metal strips are formed by an electroplating process or an electroless plating operation.

18. The method of claim 17, wherein the electroplating process or the electroless plating operation is performed in a plating bath.

19. The method of claim 16, further comprising:

removing the third patterned mask layer;

removing part of the second seed layer not covered by the second and third metal strips;

removing the second patterned mask layer; and

removing part of the first seed layer not covered by the first, second, third, and fourth metal columns.

20. The method of claim 16, wherein the first, second, and third patterned mask layers are formed by at least one of: a photolithographic process, an additive process, or a laser ablation process.

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